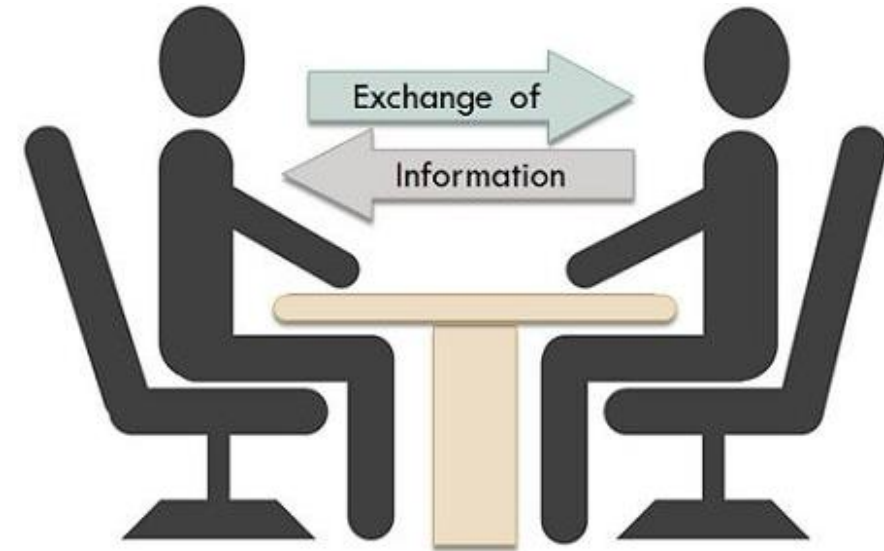
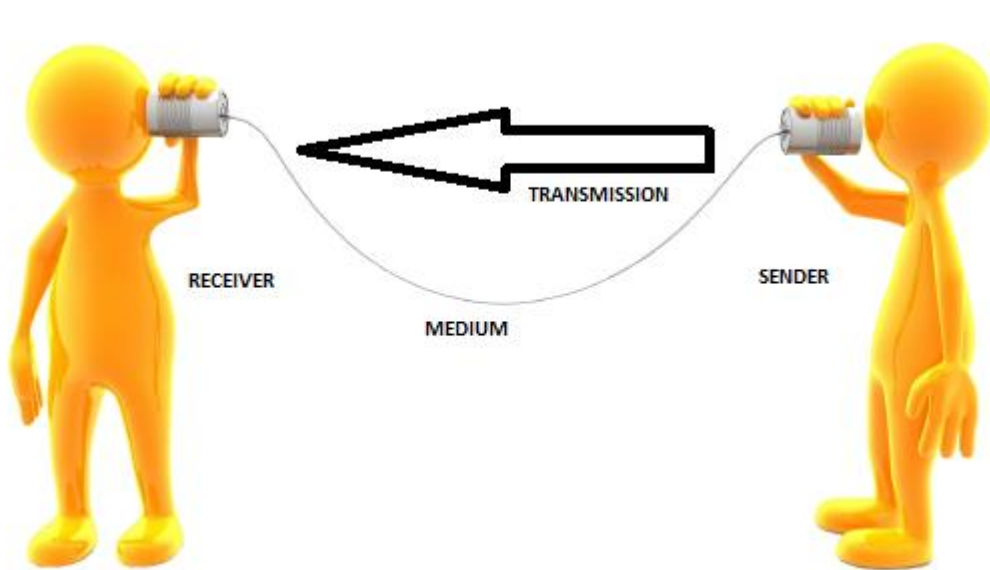


Module 5: Analog and Digital Communication

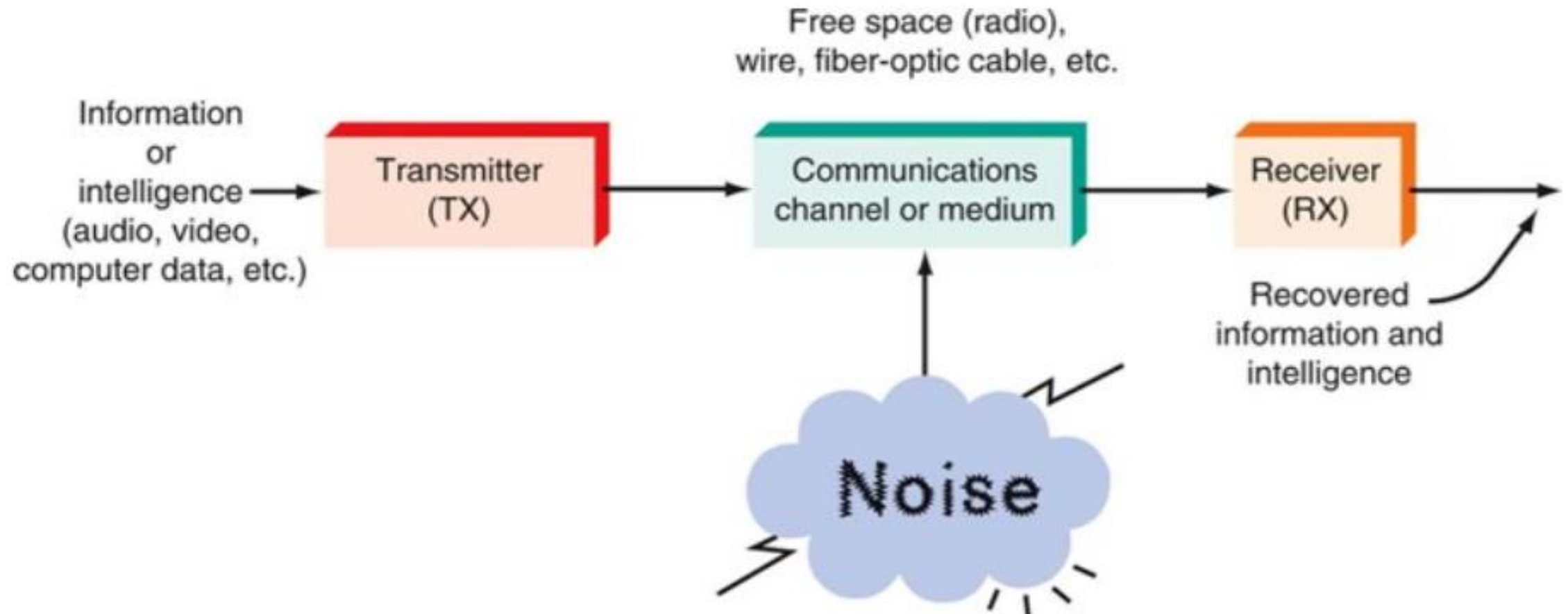
What is Communication?

Transfer of information from one point to other (or)
Exchange of Information between two points

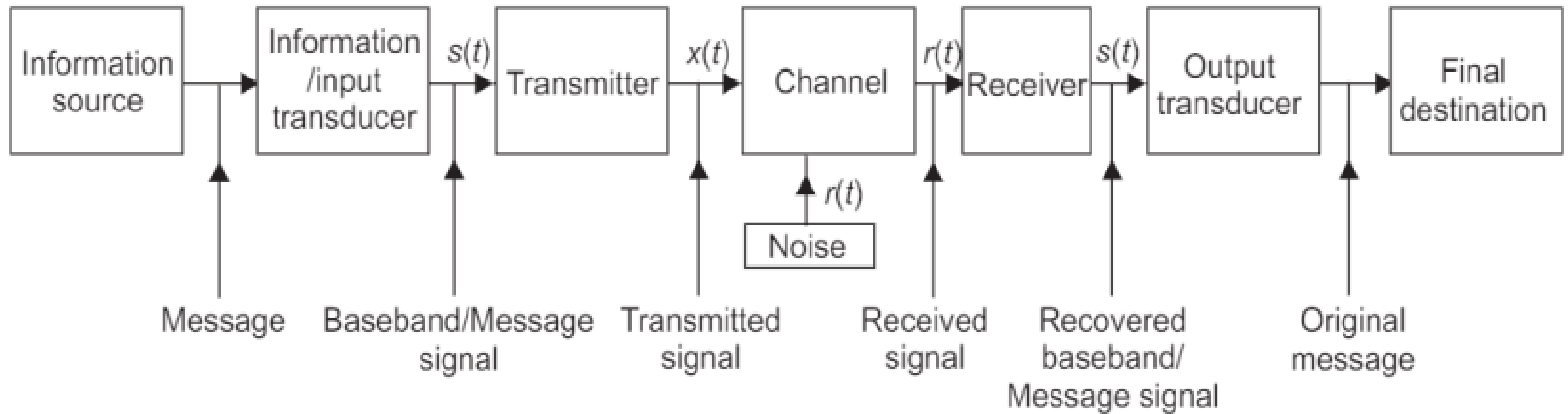


- ☐ Sending data or Information using Electronic devices: Electronic Communication
- ☐ Allows rapid transfer of goods, money and ideas

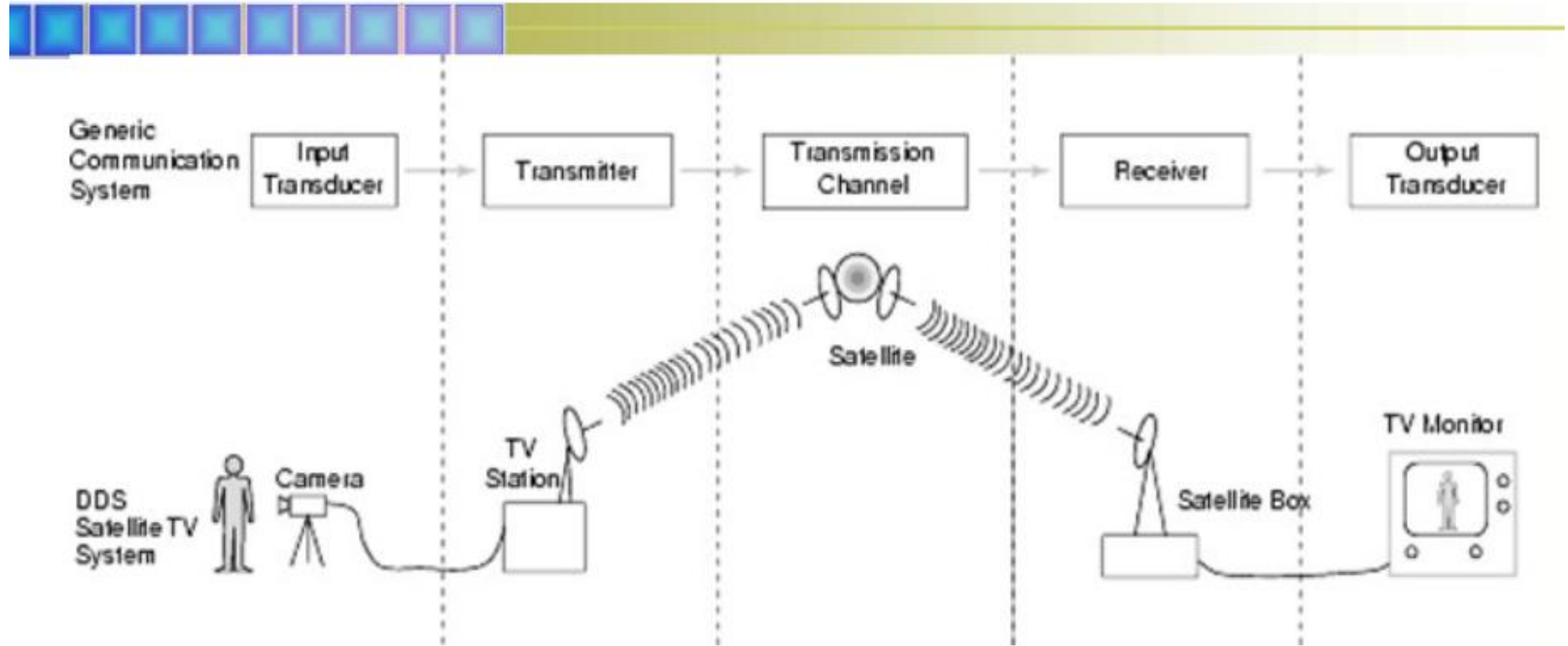
General form of a Basic Communication System



General form of a Basic Communication System



Examples of Communication



Constituents of a Communication System

- Information source and transducer
- Transmitter
- Channel or medium
- Noise
- Receiver
- Output transducer and final destination

Information Source and Transducer

- ❑ A communication system transmits information from an **information source (message)** to a **destination**.
- ❑ Examples: Voice, Live scenes(video), music, written text, and e-mail.
- ❑ A transducer is a device that **converts a non-electrical energy** into its corresponding **electrical energy** called **signal** and vice versa.

Example Transducers: Sound : Microphone

Picture: Camera

Text : Keyboard

Temperature/Pressure: Sensor with transducer

Example Transducers



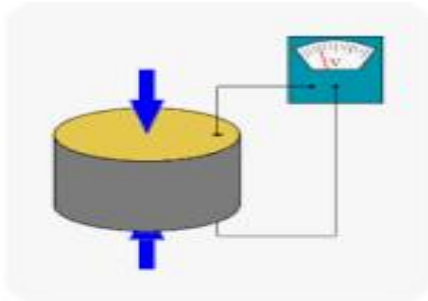
Ultrasonic Transducer



Piezoelectric Transducer



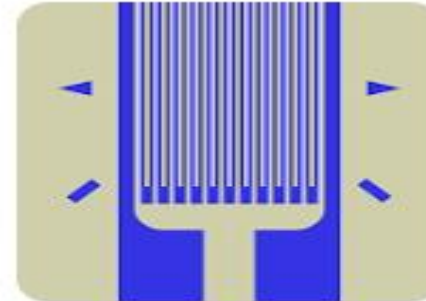
Pressure
sensor



Piezoelect...
sensor



Thermoco...



Strain gauge



Inductive
sensor

Type of Signals

Signals are functions that carry information. We use signals to convey information from place to place. In electronics, signals are mainly in the form of varying voltages

There are *two types of signals*.

(a) Analog signals

(b) Digital signals

Analog Signals

- Analog signals are continuous signals.
- The values of voltage will change in a continuous range w.r.t time.
- Usually represented using sinusoidal waves.
- Records the information as it is.
- These signals are used in analog devices
- More affected by Noise
- Examples: Any natural sound, human voice, data read by analog devices.

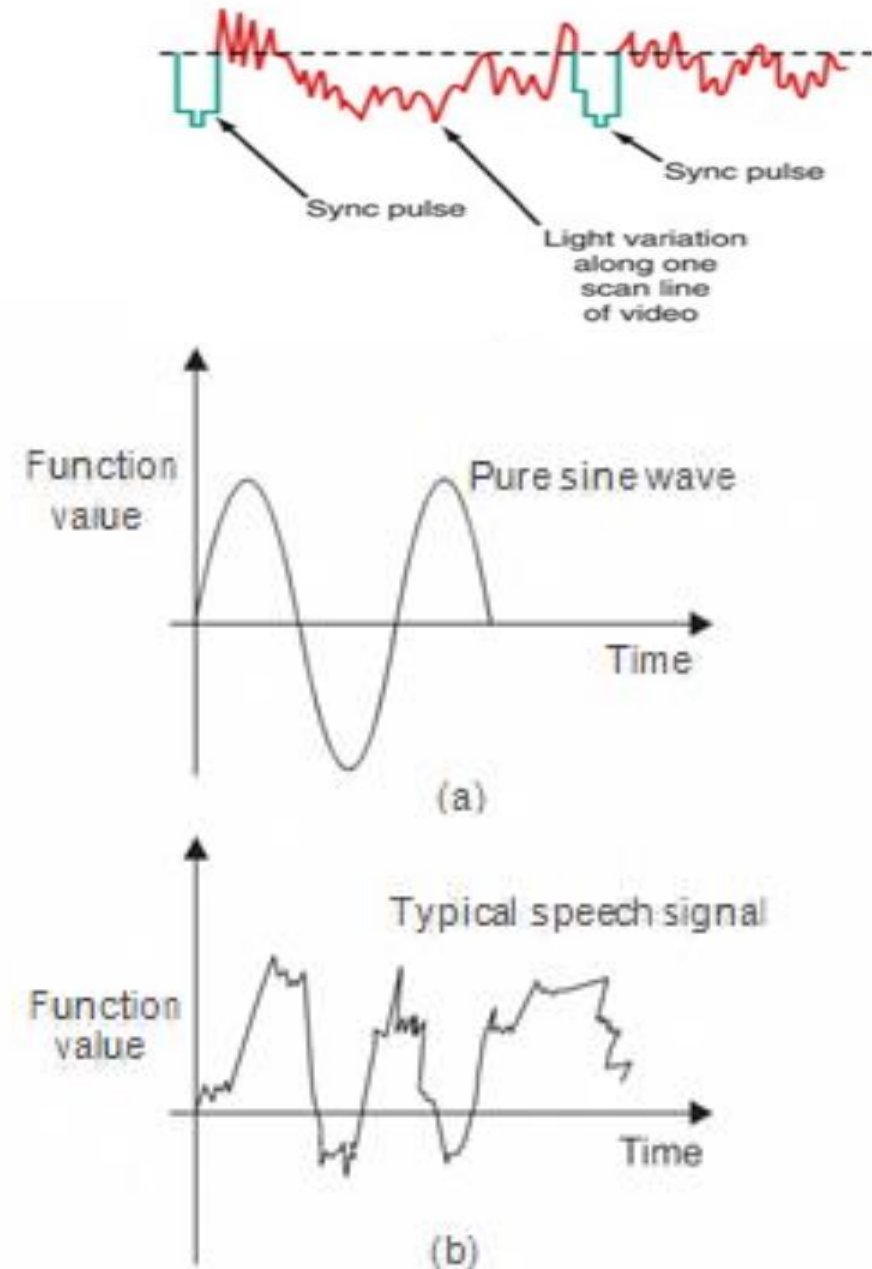
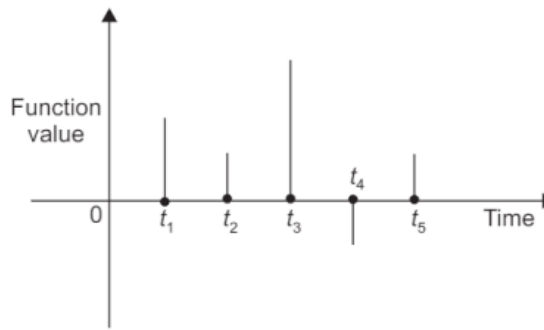
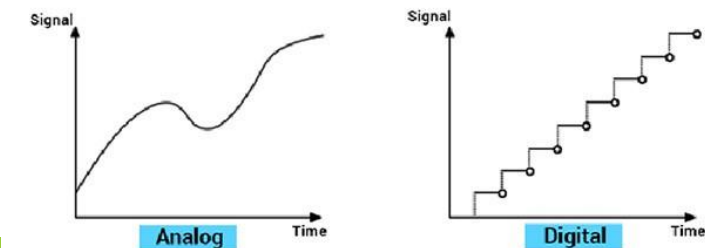
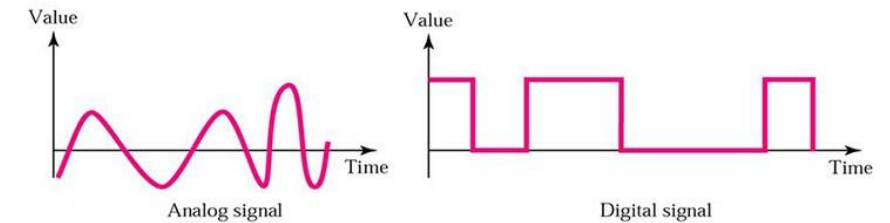
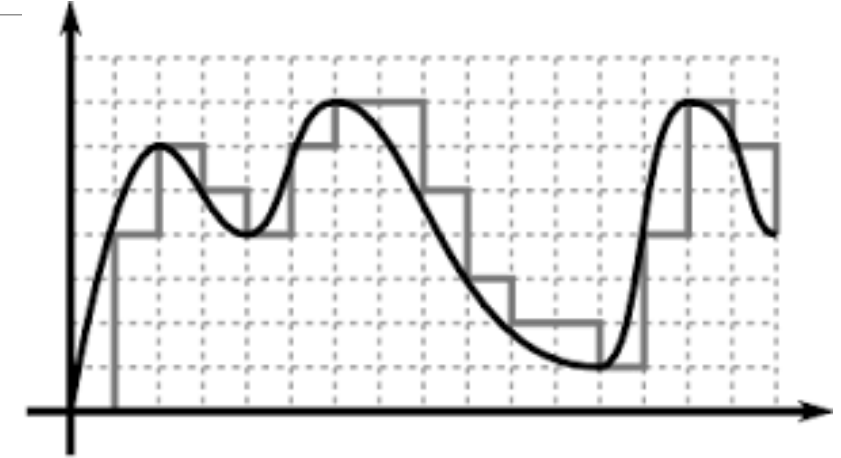
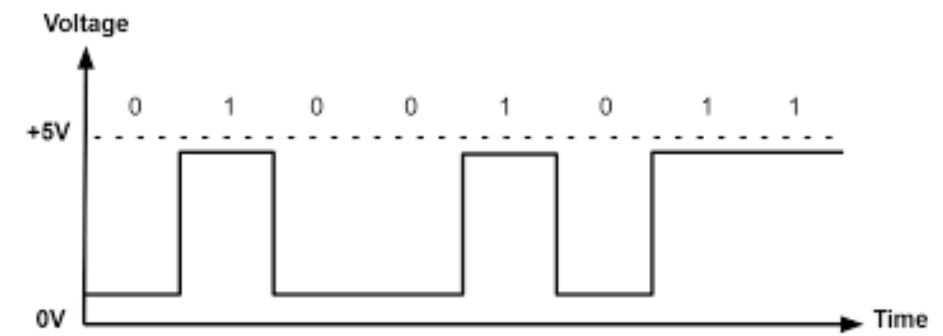


Fig. 1.2 Analog signals: (a) Pure sine wave, (b) Typical speech signal

Digital Signals



- Digital signals are not continuous, but change in discrete steps or increments they are discrete signals.
- The values of voltage will be discontinuous w.r.t time.
- Can represent in the form of square waves..
- Converts the information into binary form.
- Generated by Digital electronic devices like computers, smartphones, smartwatches, etc.
- More resilient to Noise
- **Examples: Electronic signals, computer signals, data read by digital devices.**



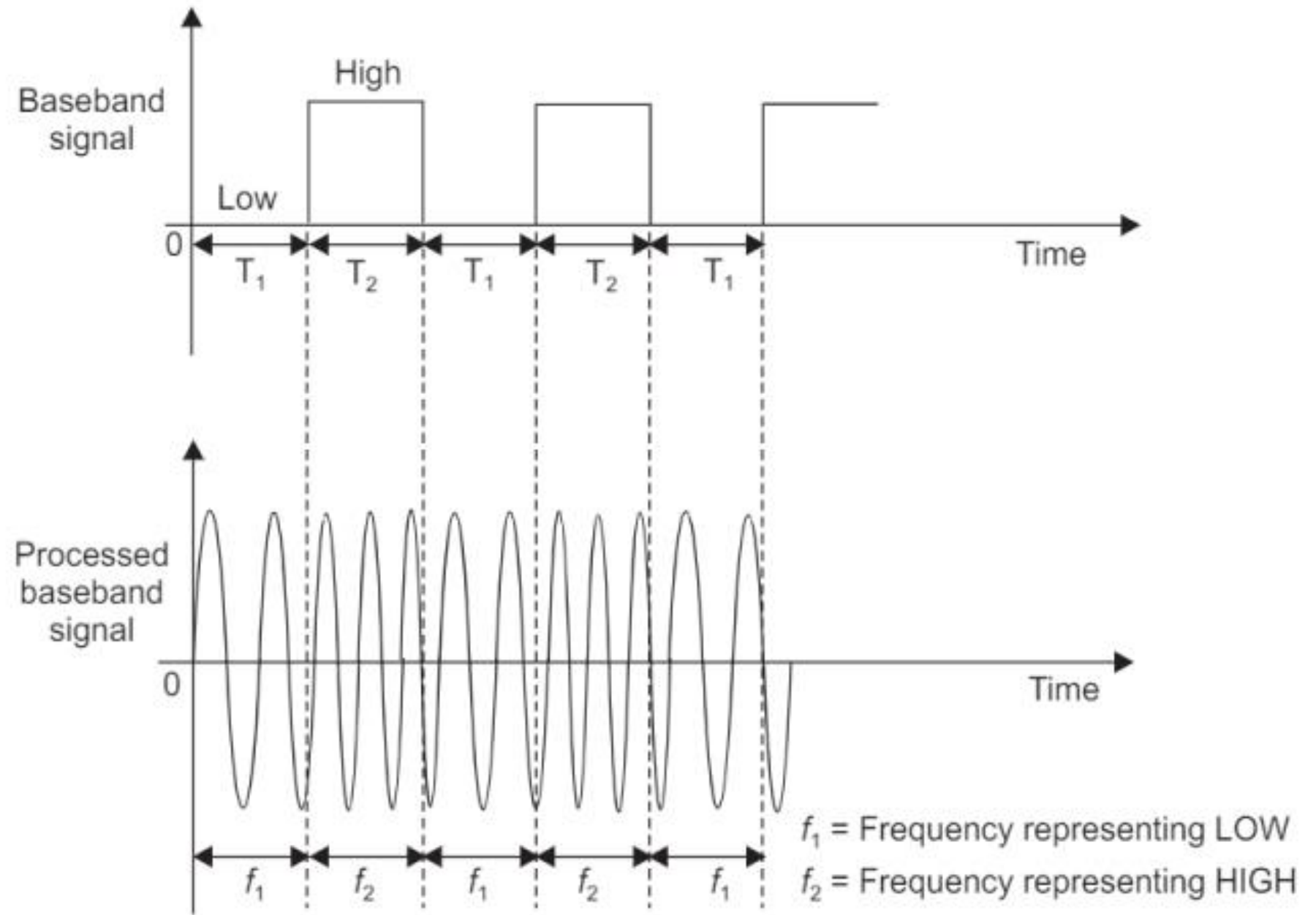
TRANSMITTER

- The **transmitter** is a collection of electronic components and circuits that converts the electrical signal into a signal suitable for transmission over a given medium.
- Transmitters are made up of oscillators, amplifiers, tuned circuits and filters, modulators, frequency mixers, frequency synthesizers, and other circuits.

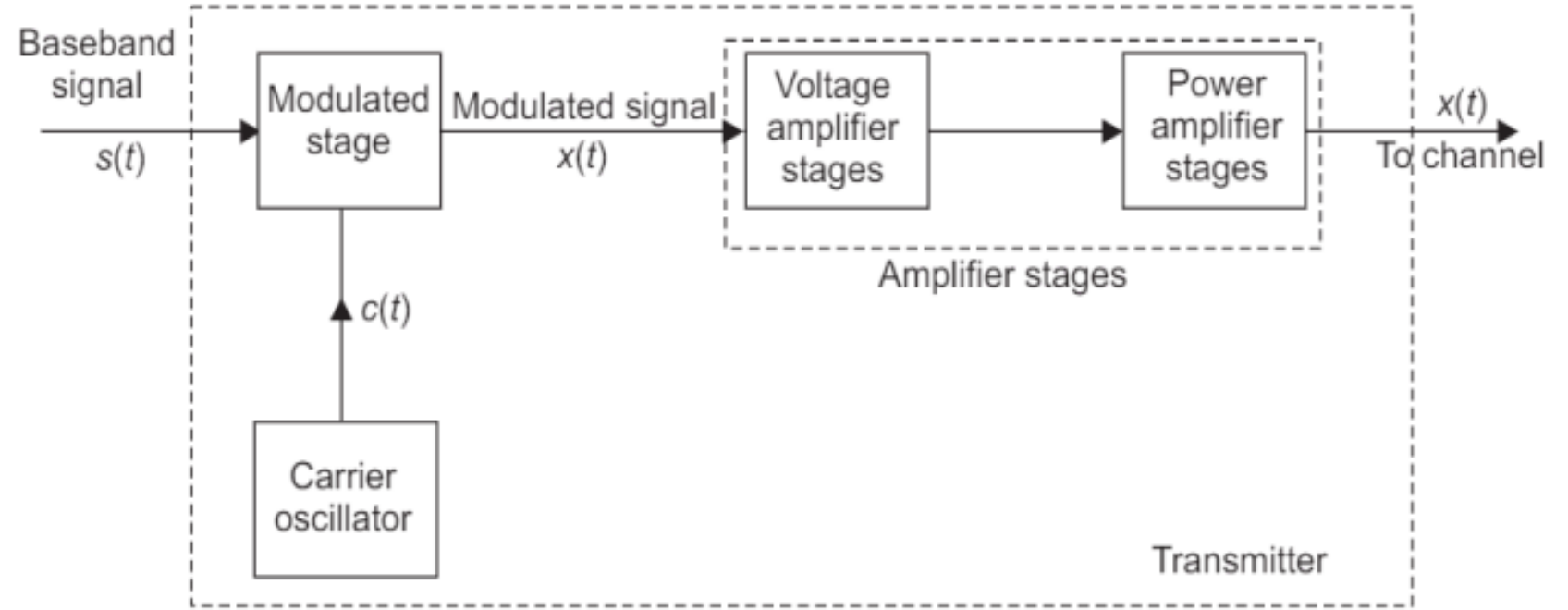
TRANSMITTER

- ❑ The base band signal, output from the input transducer, is input to the transmitter.
- ❑ The transmitter section processes the signal prior to transmission. The nature of processing depends on the type of communication system.
- ❑ There are two options for processing signals prior transmission
 - ❑ (i) The baseband signal, which lies in the low frequency spectrum, is translated to a higher frequency spectrum---**carrier communication system**
 - ❑ (ii) The baseband signal is transmitted without translating it to a higher frequency spectrum.----**baseband communication system.**

Processing of a baseband Signal



ANALOG TRANSMITTER



- $s(t)$ applied to the modulated stage, the baseband signal is translated from low to high frequency spectrum.
- This stage also receives another input called the *carrier signal*, $c(t)$, which is generated by a high frequency carrier oscillator.
- Modulation takes place at this stage with the baseband and the carrier signals.
- The baseband signal is translated to a high frequency spectrum and the carrier signal is said to be modulated by the baseband signal.
- The output of the modulated stage is called the *modulated signal*, and is designated as $x(t)$.

Power Amplifier Stage

- ❑ The *power amplifier stage* amplifies the power of the modulated signal and thus it carries enough power to reach the receiver stage of the communication system.
- ❑ Finally, the signal is passed to the transmission medium or channel.

Antenna

- For Wireless transmission of signals

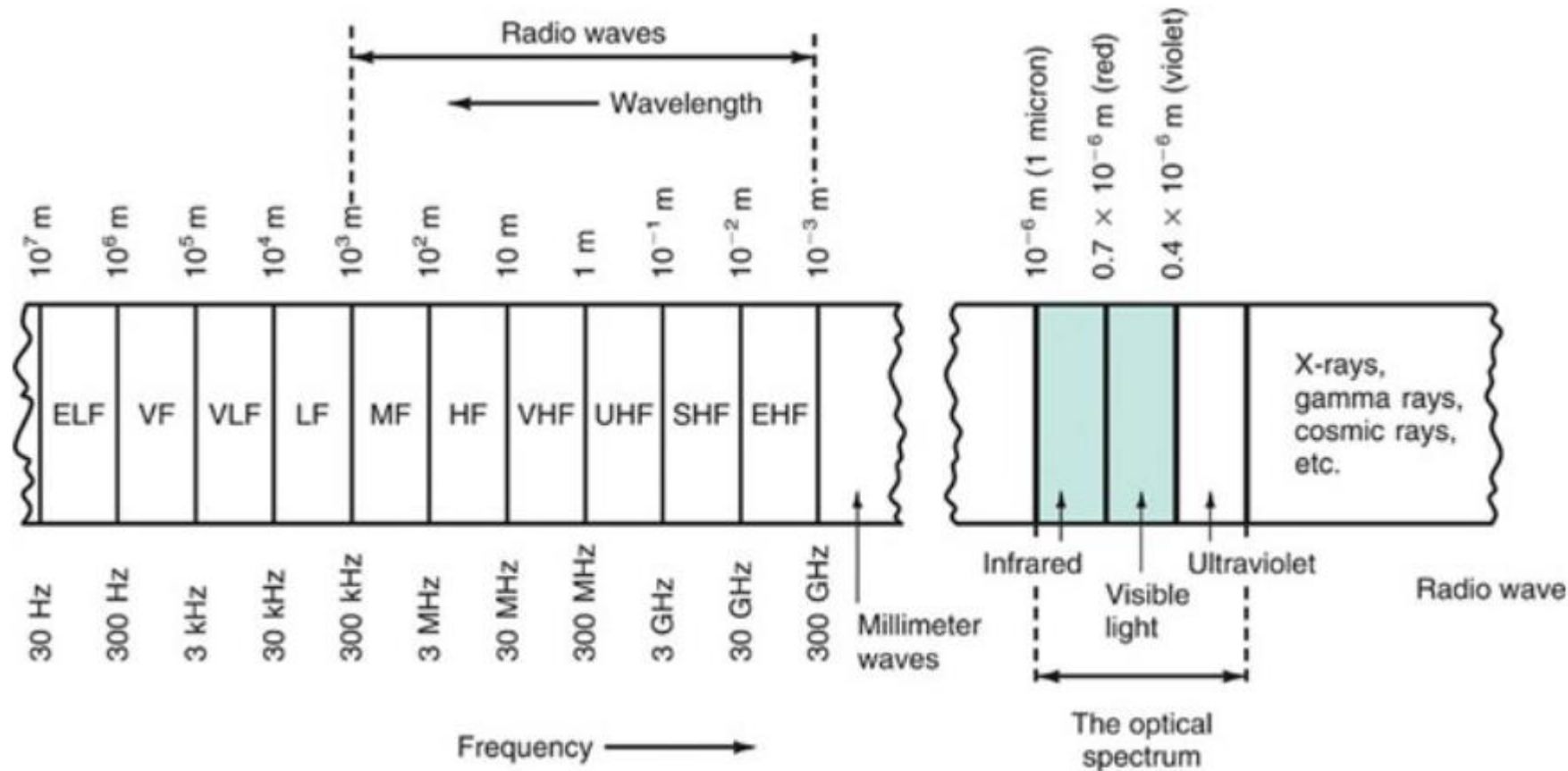
Antenna: Converts electrical signal to EM waves



Communication Channel

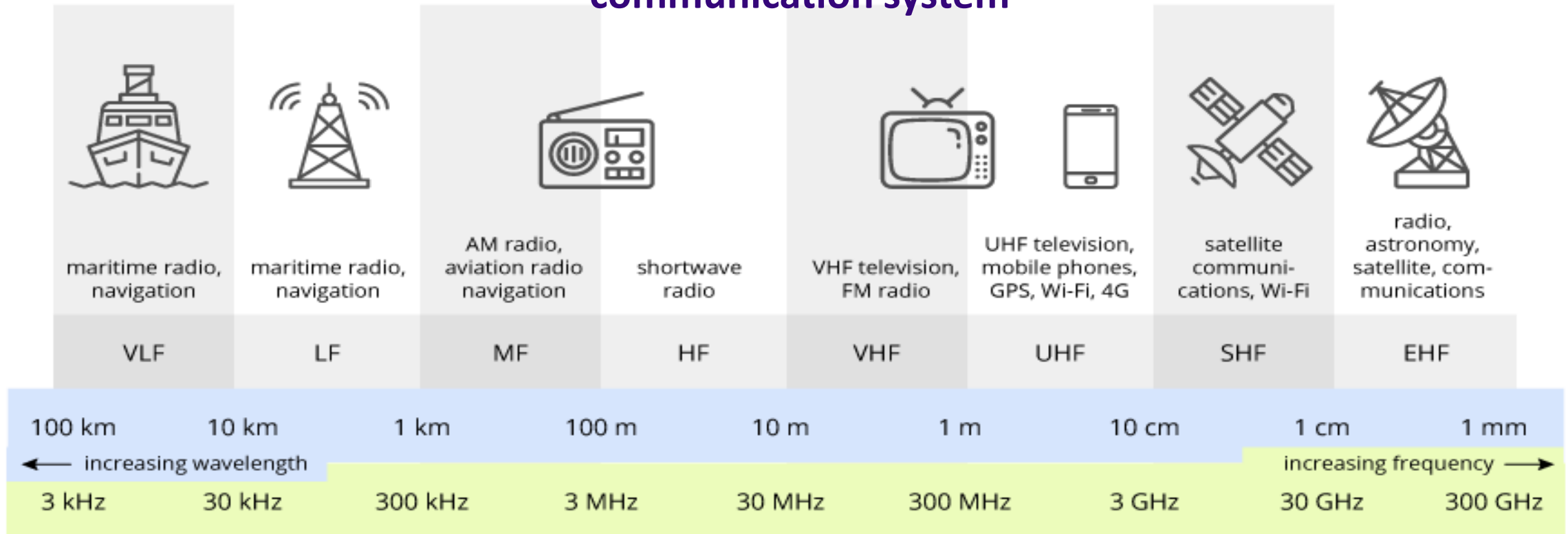
- The **communication channel** is the medium by which the electronic signal is sent from one place to another.
- Types of media include
 - Electrical conductors
 - Optical media
 - Free space
 - System-specific media (e.g., water is the medium for sonar).

Communication Bands (Electro magnetic Spectrum)



Communication Bands (Radio Frequency Spectrum)

Radio signals are transmitted as Electromagnetic waves (radio waves) in a Radio communication system



The RF Spectrum

Table 1.1: Classification of radio frequency (RF) spectrum alongwith the associated applications in communication systems.

Radio frequency range	Wavelength (meters)	Class	Applications
10–30 kHz	$3 \times 10^4 - 10^4$	Very Low Frequency (VLF)	Point-to-point communication (long distance)
30–300 kHz	$10^4 - 10^3$	Low Frequency (LF)	Point-to-point communication (long distance) and navigation
300–3000 kHz	$10^3 - 10^2$	Medium Frequency (MF)	Radio broadcasting
3–30 MHz	$10^2 - 10$	High Frequency (HF)	Overseas radio broadcasting, Point-to-point radio telegraphy, and telephony
30–300 MHz	$10 - 1.0$	Very High Frequency (VHF)	FM broadcast, television, and radar
300–3000 MHz	$1.0 - 0.1$	Ultra High Frequency (UHF)	Television and navigation
3000–30,000 MHz	$0.1 - 0.01$	Super High Frequency (SHF)	Radar navigation and radio relays

CHANNEL OR MEDIUM

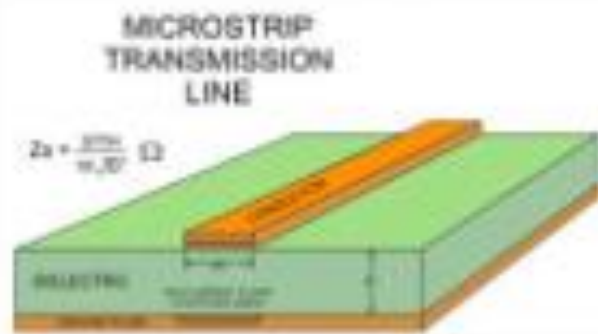
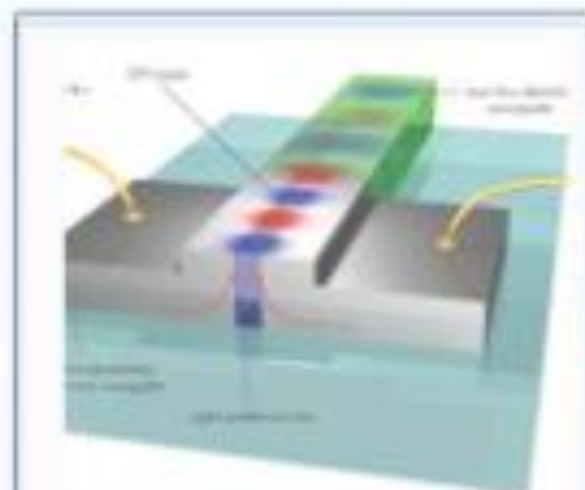
- ❑ The transmission medium between the transmitter and the receiver is called a **channel**.
- ❑ The transmitted signal should have adequate power to withstand the **channel noise**.
- ❑ The channel characteristics also impose constraints on the **bandwidth**
- ❑ Depending on the physical implementations, one can classify the channels in the following two groups:
 - ❑ **a) Hardwired channels and**
 - ❑ **b) Softwired channels.**

Hardwired (Hardware) Channels

Are manmade structures which can be used as transmission medium. There are following three possible implementations of the hardware channels.

- ❑ Transmission lines
- ❑ Waveguides
- ❑ Optical Fiber Cables (OFC)

Transmission lines and waveguides



Soft-wired (Software) Channels

- ❑ Natural resources which can be used as the transmission medium for signals.
- ❑ Eg. Air or Open space and Sea water.
- ❑ The signals are transmitted in the form of electromagnetic (EM) waves also called radio waves.
- ❑ Radio waves travel through open space at a speed equal to that of light ($c = 3 \times 10^8$ m/s)

NOISE

- ❑ Noise is defined as unwanted electrical energy of random and unpredictable nature .
- ❑ Noise is a highly undesirable part of a communication system, and has to be minimized.
- ❑ When noise is mixed with the transmitted signal, it rides over it and deteriorates its waveform
- ❑ SNR and Noise figure (F)

Measurement of SNR

Now, if P_s is signal power and P_n is noise power, then SNR expressed as S/N, is given as

$$\frac{S}{N} = \frac{P_s}{P_n}$$

If $P_s = V_s^2 R$ and $P_n = V_n^2 R$, then

$$\frac{S}{R} = \frac{P_s}{P_n} = \frac{V_s^2 R}{V_n^2 R} \quad \dots(2)$$

where V_s is signal voltage and V_n is noise voltage.

In addition, it is assumed that both the signal and noise powers are dissipated in the same resistor R . Therefore, SNR can be expressed in terms of decibels (dB) as

$$\left(\frac{S}{N}\right)_{dB} = 10 \log_{10} \left(\frac{V_s^2}{V_n^2} \right)$$

$$\left(\frac{S}{N}\right)_{dB} = 20 \log_{10} \left(\frac{V_s}{V_n} \right) \quad \dots(3)$$

Noise Factor and Noise Figure

$$F = \frac{\frac{S}{N} \text{ Power at the input terminals of the circuit}}{\frac{S}{N} \text{ Power at the output terminals of the circuit}}$$

Noise Calculation

- SNR is ratio of signal power, S to noise power, N .

$$SNR = 10 \log \frac{S}{N} \text{ dB}$$

- Noise Factor, F

$$F = \frac{S_i/N_i}{S_o/N_o}$$

- Noise Figure, NF

$$\begin{aligned} NF &= 10 \log F \\ &= 10 \log \frac{S_i/N_i}{S_o/N_o} \text{ (dB)} \end{aligned}$$

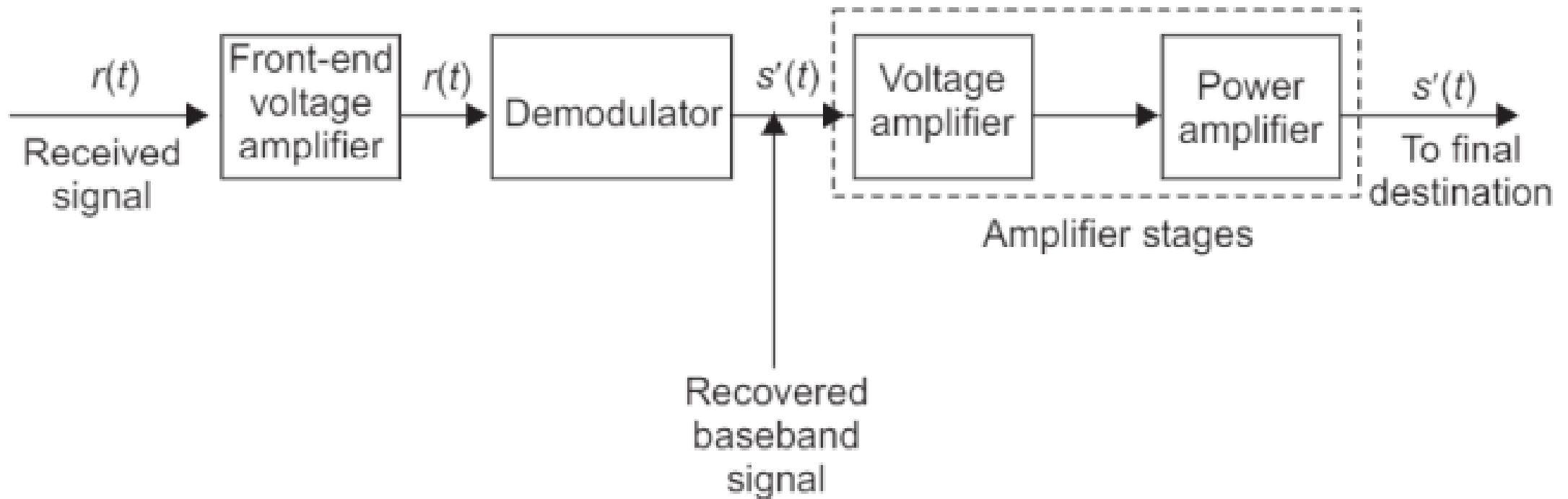
RECEIVER

- ❑ The task of the receiver is to provide the original information to the user.
- ❑ The signal received by the receiver is $r(t)$.
- ❑ This signal contains both the transmitted signal, $x(t)$, and the noise, $n(t)$, added to it during transmission

RECEIVER

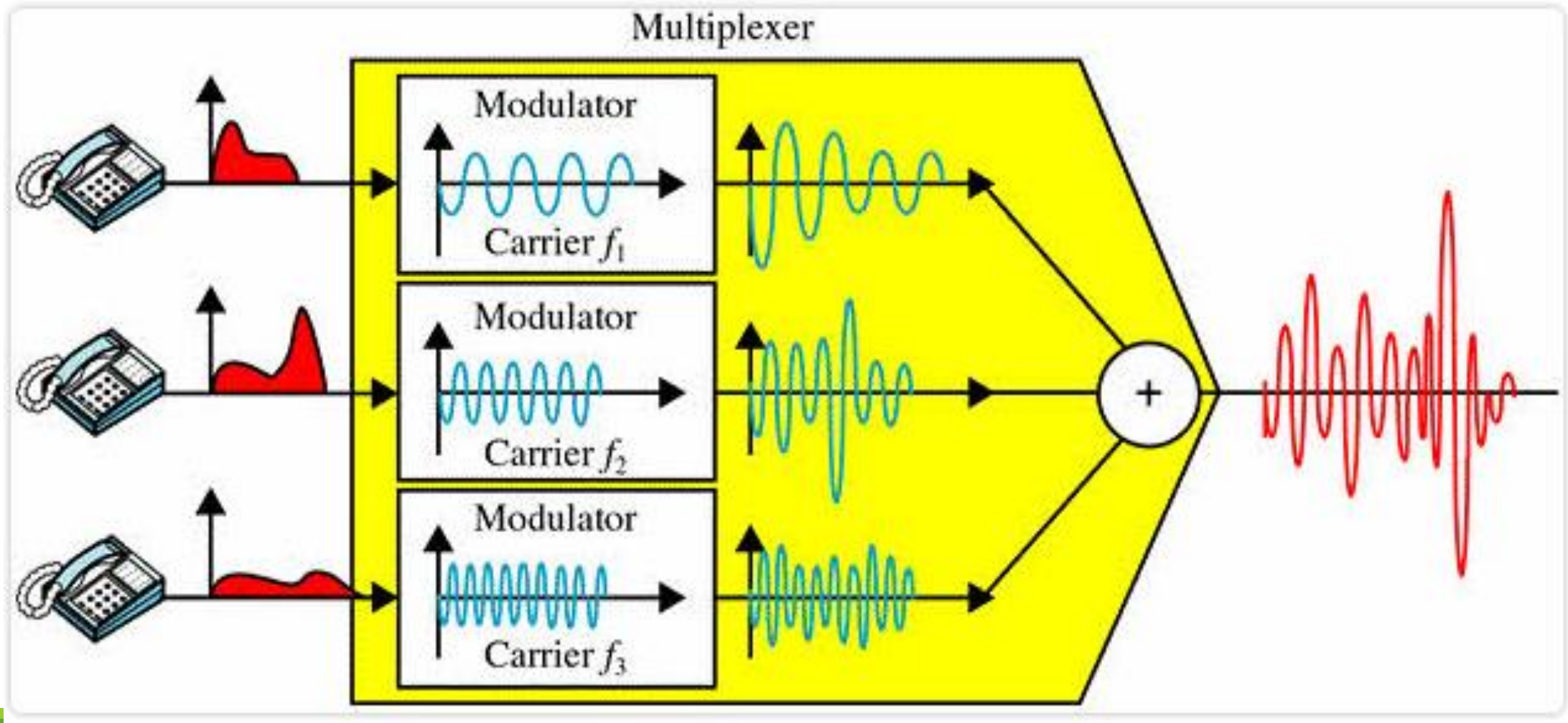
- A **receiver** is a collection of electronic components and circuits that accepts the transmitted message from the channel and converts it back into a form understandable by humans.
- Receivers contain amplifiers, oscillators, mixers, tuned circuits and filters, and a **demodulator** or detector that recovers the original intelligence signal from the modulated carrier.

Detailed block diagram of a Typical Receive Section

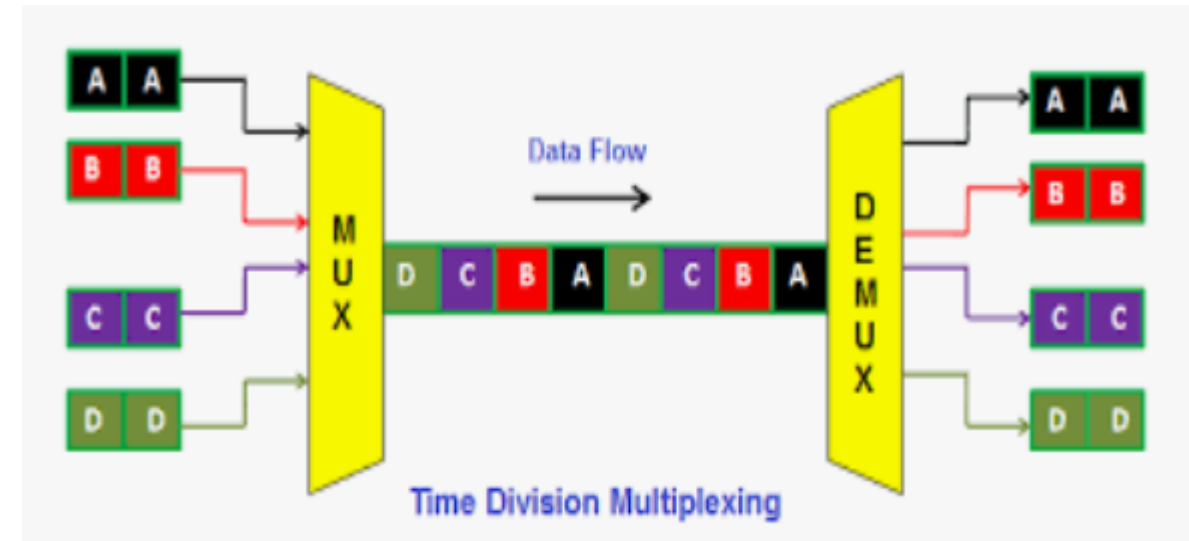
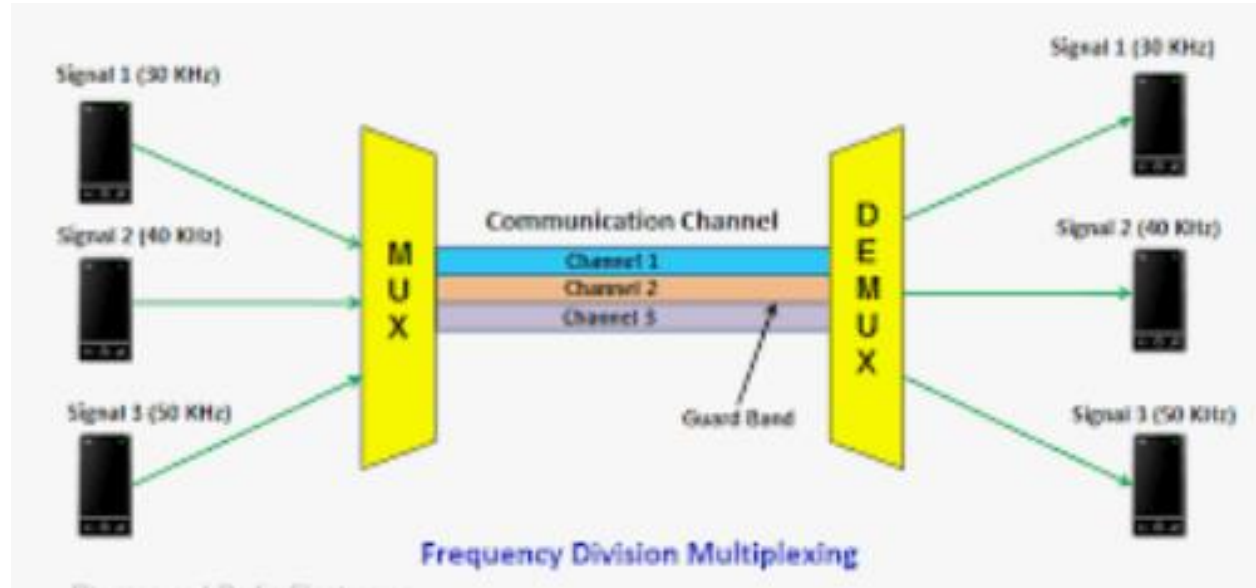


MULTIPLEXING

Multiplexing is a process which allows more than one signal to be transmitted through a single channel.



TYPES OF MULTIPLEXING



Advantages of Multiplexing

- ❑ Multiplexing allows the maximum possible utilization of the available bandwidth of the system.
- ❑ The use of multiplexing also makes the communication system economical because more than one signal can be transmitted through a single channel.

TYPES OF COMMUNICATION SYSTEMS

One may categorize communication systems based on their **physical infrastructure and the specifications of the signals** they transmit.

- ☐ **Communication Systems based on Physical Infrastructure**

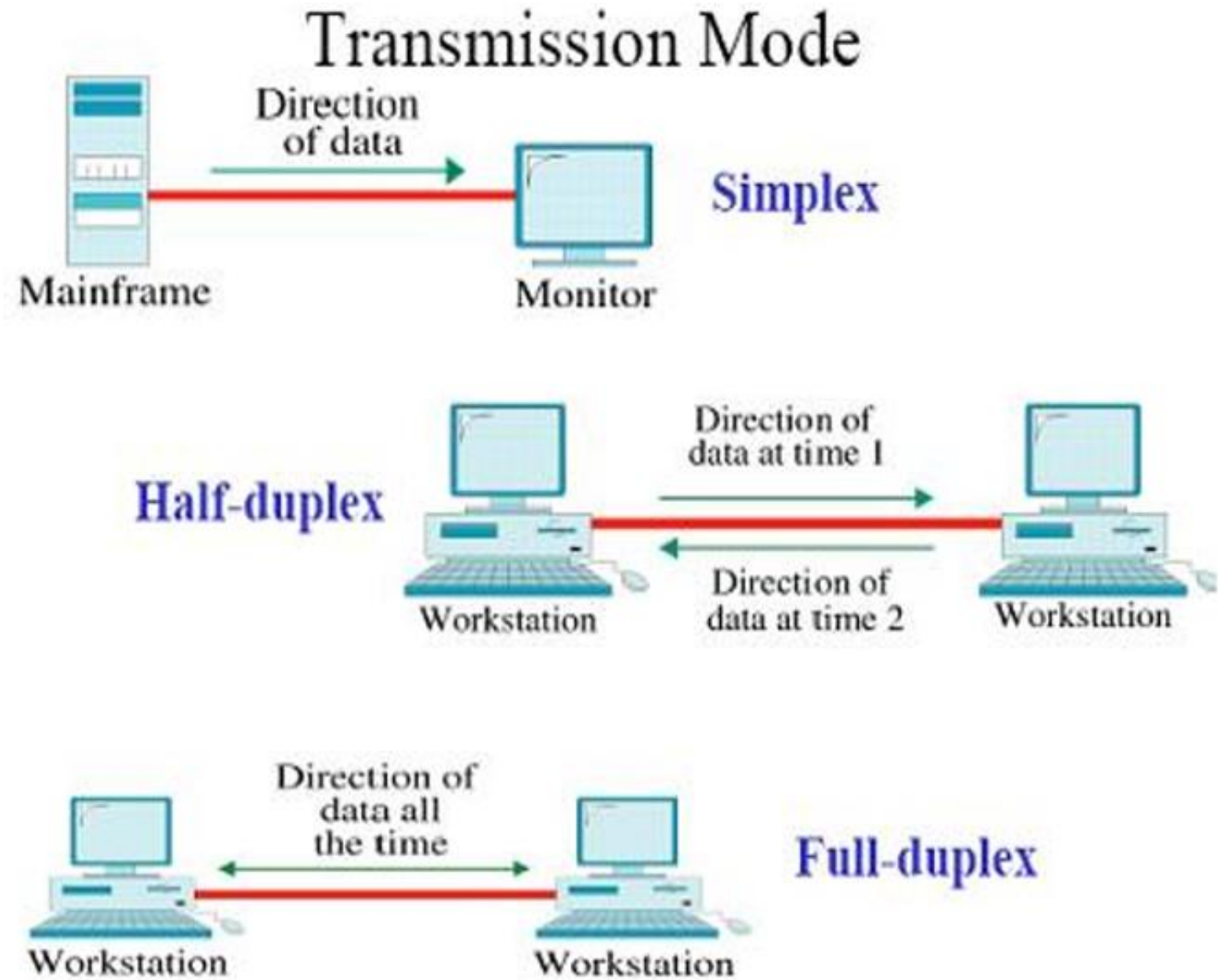
- ☐ **Line Communication Systems**

- ☐ **Communication systems based on Signal specifications**

- ☐ **Analog/ Digital Communication systems**

- ☐ **Baseband/ Carrier Communication systems**

1.Communication Systems based on Physical Infrastructure



2. Communication Systems based on Signal Specifications

The signal specifications used to decide the type of communication include:

- ❑ Nature of baseband or information signal
- ❑ Nature of the transmitted signal.

- ❑ A. Based on the **nature of the baseband signal**
 - ❑ Analog communication systems
 - ❑ Digital communication systems.

B. Based on the **nature of the transmitted signal**

❑ The two systems can then be put under following categories:

- ❑ Baseband communication system
- ❑ Carrier communication system

Thus, there are **four types** of communication system categories based on signal specification. These are:

- ❑ Analog communication system
- ❑ Digital communication system
- ❑ Baseband communication system
- ❑ Carrier communication system

Of the four, at least two types are required to specify a particular communication system. These groups can be put as:

Analog/Digital communication system

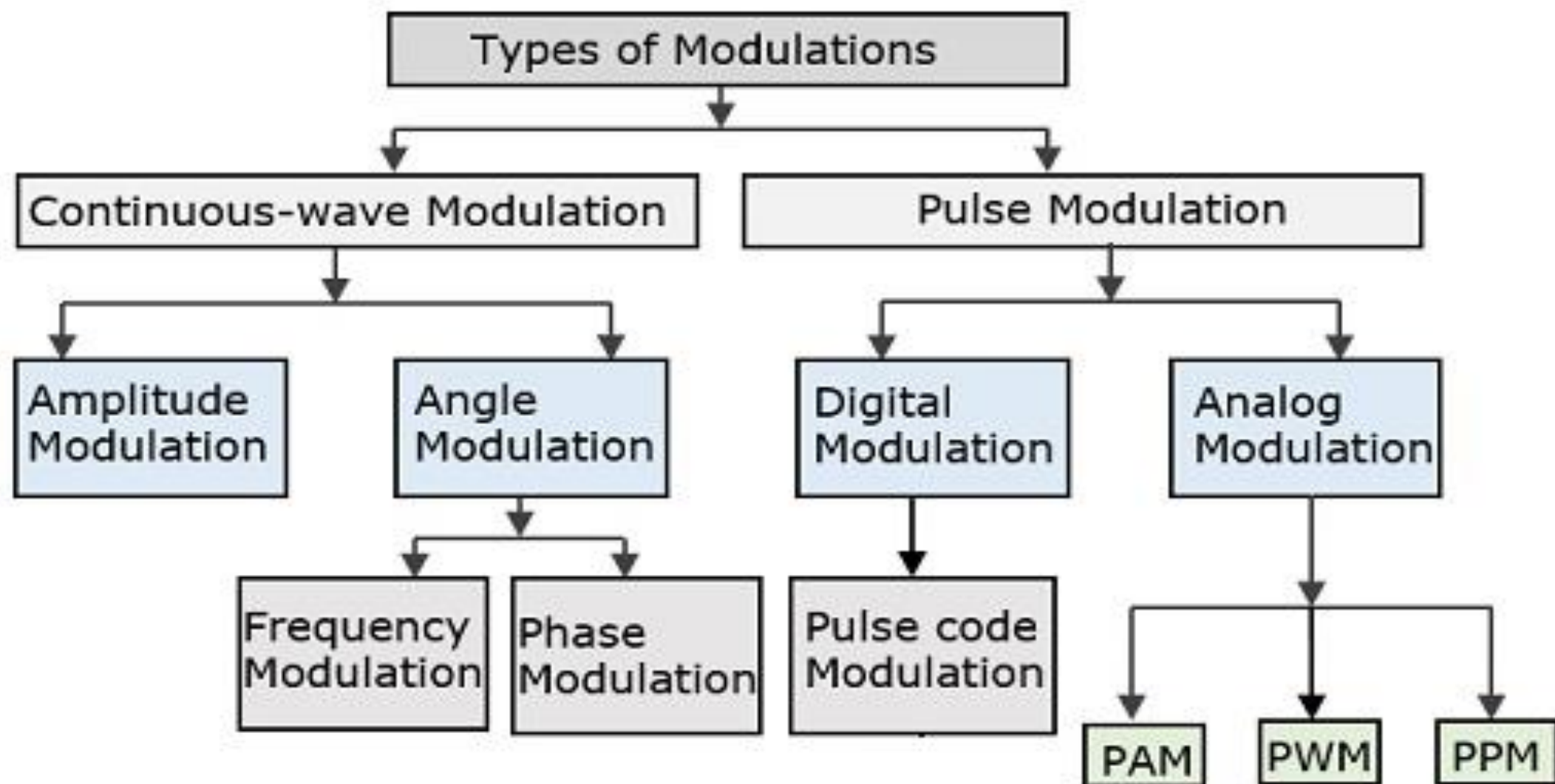
Baseband/Carrier communication system

Modulation

- ❑ Modulation – process of translating the low frequency baseband signal to higher frequency spectrum
 - ❑ Process of changing the parameters of the carrier signal, in accordance with the instantaneous values of the modulating signal.
 - ❑ Need for modulation: :
- <https://www.youtube.com/watch?v=00ZbuhPruJw>

Need for Modulation

- Improves Quality of reception
- Reduces Height of antenna
- Options for Multiplexing
- Bandwidth Extension
- Increased Range of Communication
- Reduced noise and interference

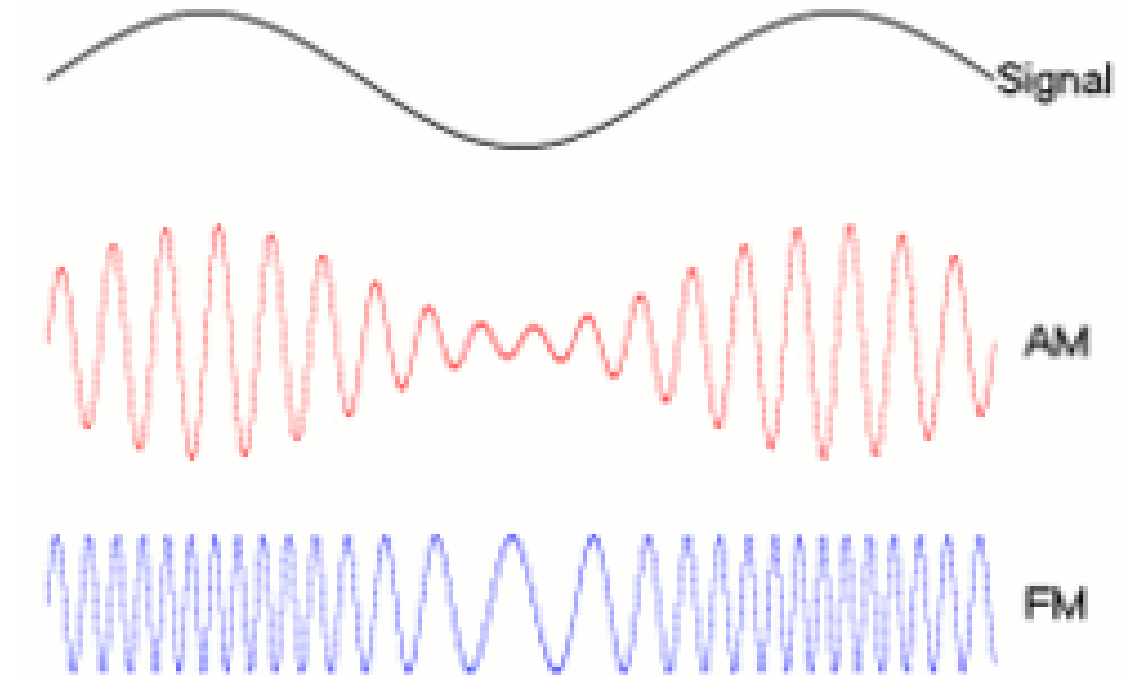


❑ Types of Analog (Continuous Wave) Modulation

❑ Amplitude modulation

❑ Frequency modulation

❑ Phase modulation



1. Amplitude Modulation

❑ **Amplitude modulation (AM)** -modulation technique in which the instantaneous amplitude of the carrier signal is varied in accordance with the instantaneous amplitude of the analog modulating signal to be transmitted

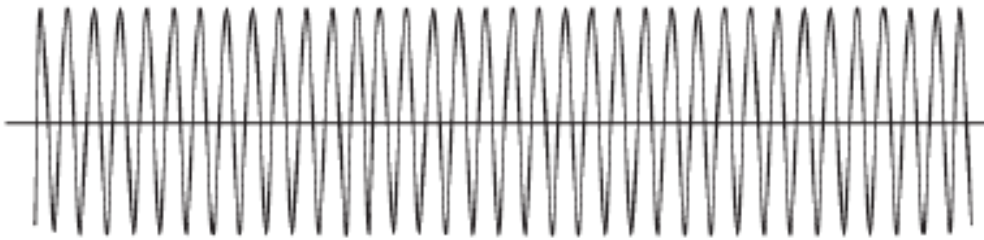
❑ **Modulating signal** - an analog baseband signal which is random and has a low frequency

❑ **Carrier signal**- a sinusoidal wave with high frequency

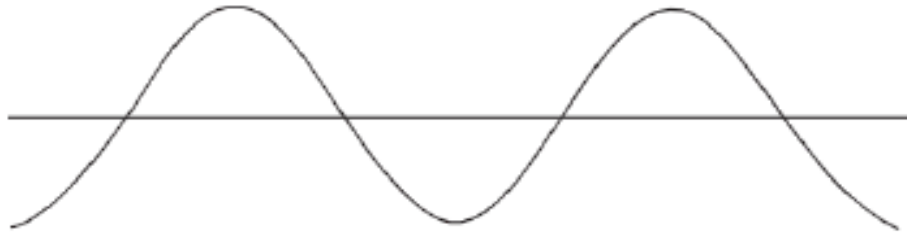
Variations in amplitude of carrier signal represent the information

Amplitude Modulation

High frequency carrier



Modulating signal



Modulated signal



- **Note :** It can be observed from the figure that the amplitude of the carrier wave is varied in accordance with the modulating signal while the frequency and phase of the carrier signal remains unchanged
- Modulating signal seems to be superimposed on the carrier signal
- Amplitude variations in the peak values of the carrier signal exactly replicates the modulating signal at different points of time which is known as an envelope.
- **Modulation Index** $\mu = A_m/A_c$

Demonstrations

<https://demonstrations.wolfram.com/AmplitudeModulation/>

<https://demonstrations.wolfram.com/PowerEfficiencyOfAmplitudeModulation/>

2. Frequency Modulation

- ❑ Process of changing the frequency of the carrier signal in accordance with the instantaneous value of the modulating voltage while keeping the amplitude and phase of the carrier constant.
- ❑ The original frequency of the carrier signal is called the centre or resting frequency denoted as f_c .
- ❑ Frequency deviation (Δf) -The amount by which the frequency of the carrier wave changes or shifts above or below the resting frequency.



$$\Delta f \propto m(t)$$

Frequency Modulation

❑ The total variation of frequency of FM wave from the lowest to highest is termed as carrier swing (CS)

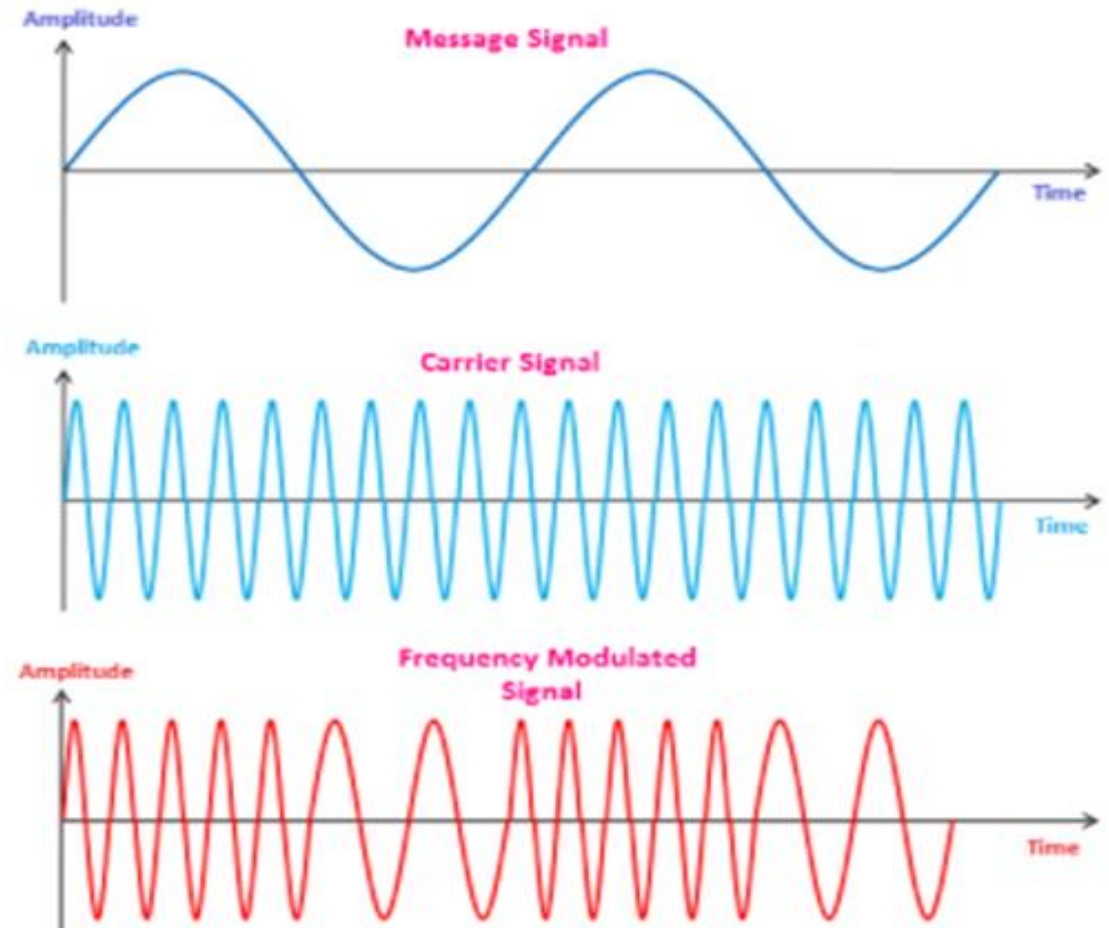
❑ $CS = 2 \times \Delta f$

❑ Modulation Index

❑ $\mu_f = \frac{\text{Frequency deviation}}{\text{Modulating frequency}} = \frac{\Delta f}{f_m}$

❑ FM Demo:

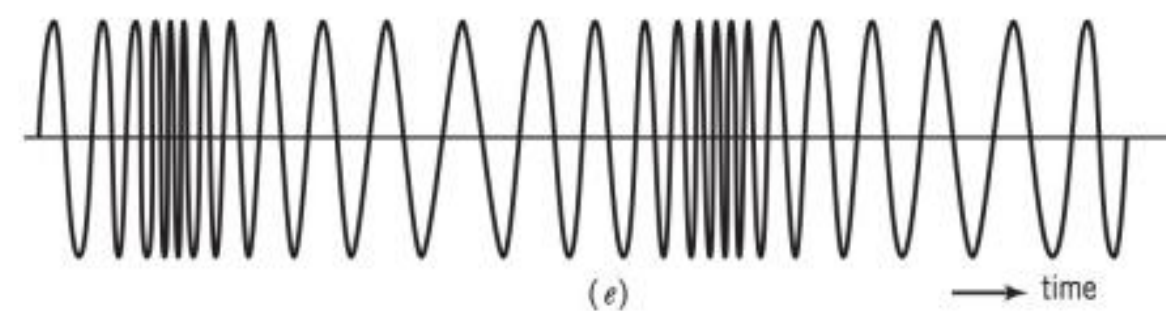
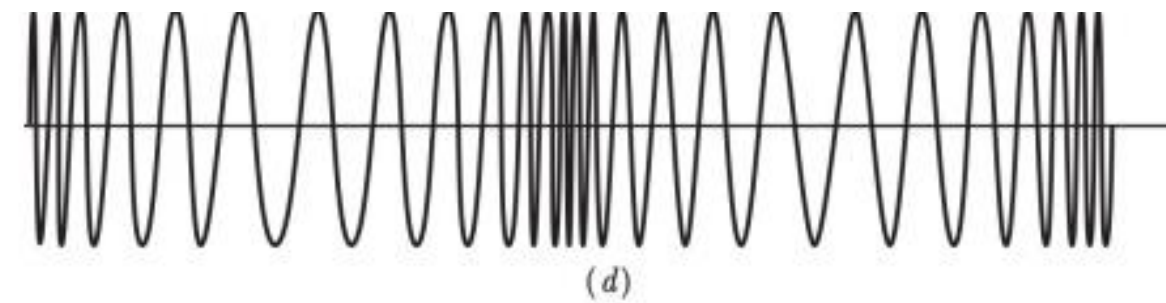
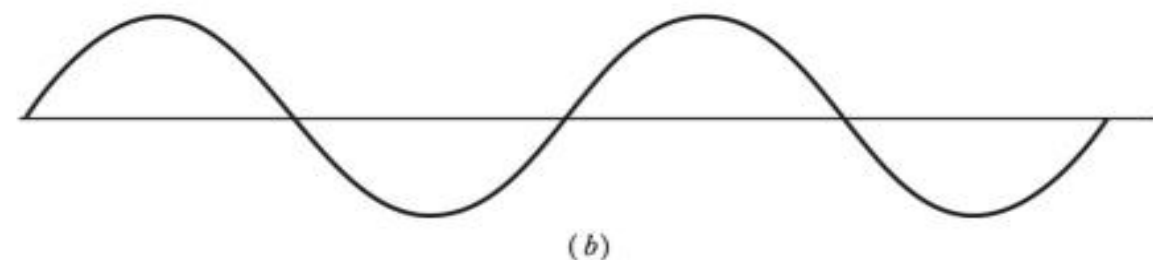
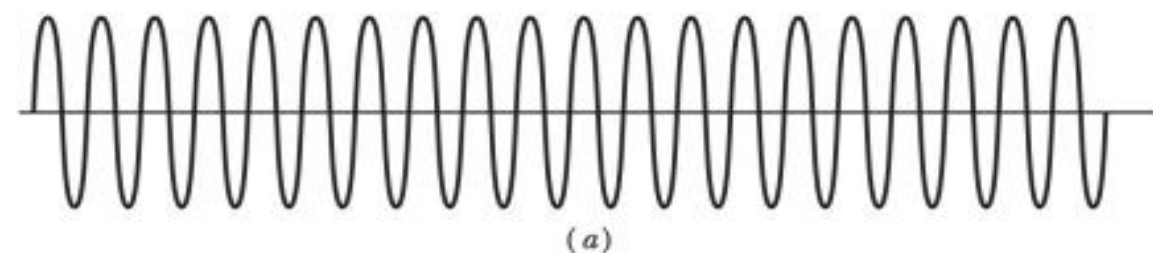
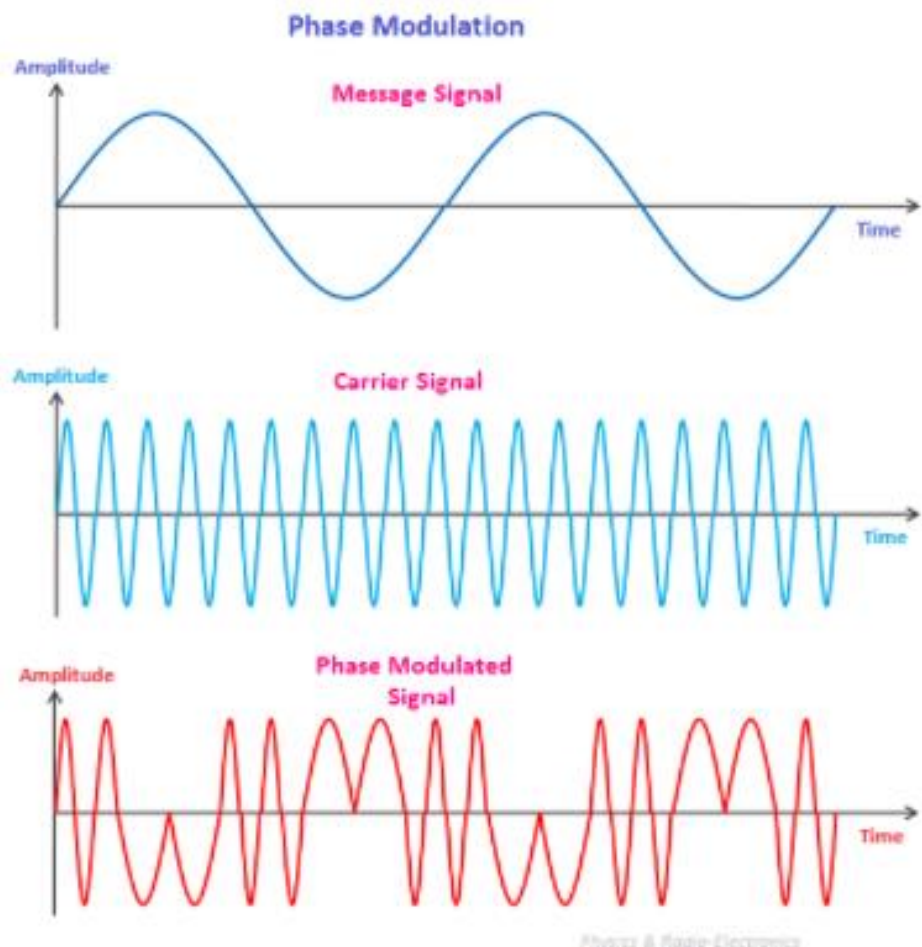
<https://youtu.be/4SOGF4AUIY>



Phase Modulation

- ❑ Process in which the instantaneous phase of the carrier signal is varied in accordance with the instantaneous amplitude of the modulating signal
- ❑ The modulating signal is mapped to the carrier signal in the form of variations in the instantaneous phase of the carrier

Phase Modulation



WHY DIGITIZE ANALOG SOURCES?

- ❑ Less sensitive to noise.
- ❑ It is easier to integrate different services
 - ❑ video and the accompanying soundtrack, into the same transmission scheme.
- ❑ The transmission scheme can be relatively independent of the source.
- ❑ Circuitry for handling digital signals is easier to repeat
- ❑ Digital circuits are less sensitive to physical effects
 - ❑ such as vibration and temperature.
- ❑ Digital signals are simpler to characterize ,this makes the associated hardware easier to design.

Contd..

☐ Easy to Implement techniques like

- ☐ Multiplexing
- ☐ Channel compensation
- ☐ Equalization
- ☐ Error correction

☐ There are techniques for removing redundancy from a digital transmission, so as to minimize the amount of information that has to be transmitted.

☐ Digital techniques make it easier to specify complex standards that may be shared on a worldwide basis.

- ☐ This allows the development of communication components with different features and their interoperation with a different component produced by a different manufacturer.

Analog to Digital Conversion

- The *Analog-to-digital Converter* (ADC) performs three functions:

Sampling

Makes the signal discrete in time.

If the analog input has a bandwidth of W Hz, then the *minimum sample frequency* such that the signal can be reconstructed without distortion is $2W$ Hz

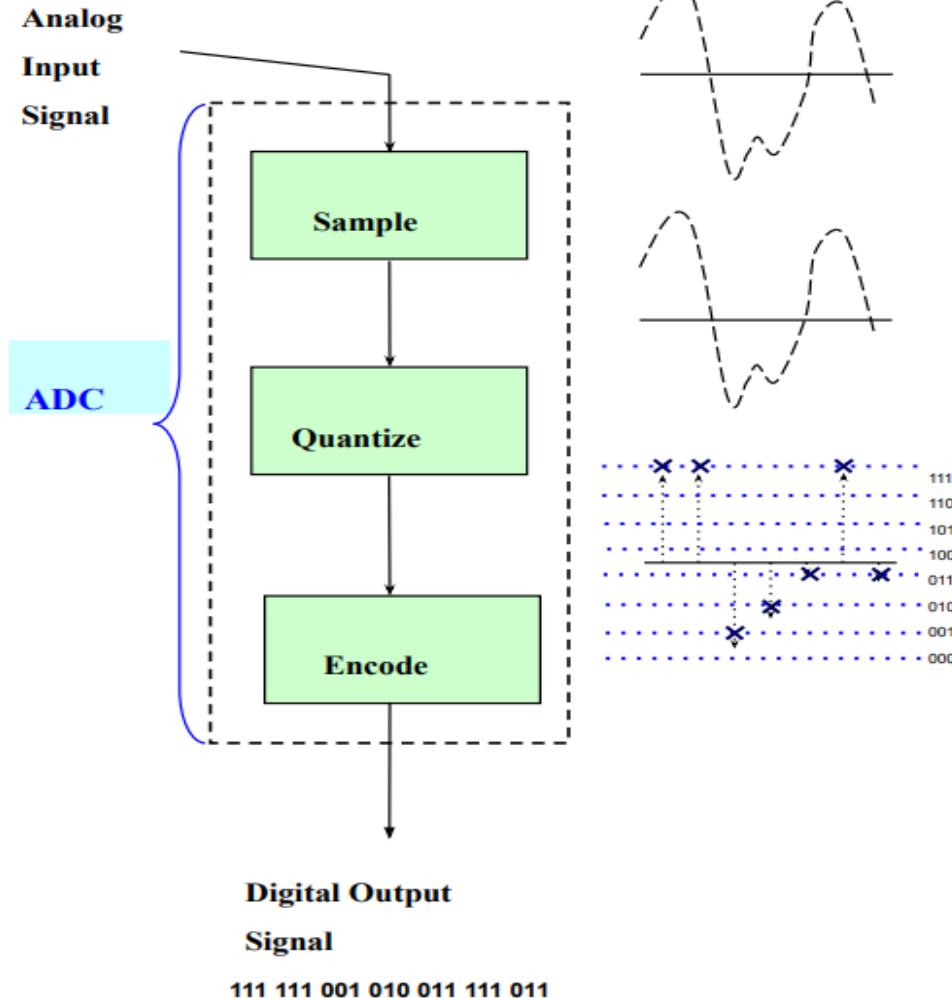
Quantization

Makes the signal discrete in amplitude.
Round off to one of q discrete levels.

Encode

Maps the quantized values to digital words that are v bits long.

- If the (Nyquist) *Sampling Theorem* is satisfied, then only quantization introduces distortion to the system.



Pulse Modulation

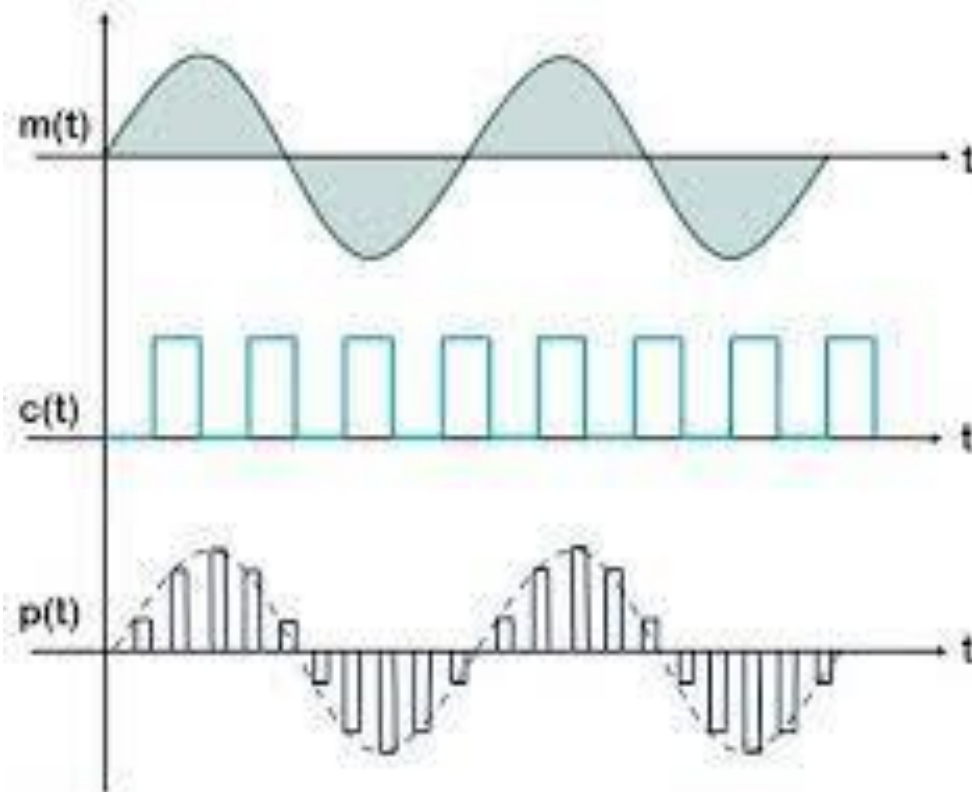
- ❑ Pulse Modulation - Used to transmit analog information such as continuous speech or data
- ❑ Has the advantage of ability to use constant amplitude pulses

Pulse Modulation Schemes

Two types of pulse modulation schemes

1. Pulse Amplitude Modulation(**PAM**)
2. Pulse Time Modulation(**PTM**)
 - Pulse Width Modulation (PWM)
 - Pulse Position Modulation (PPM)

Pulse Amplitude Modulation(PAM)



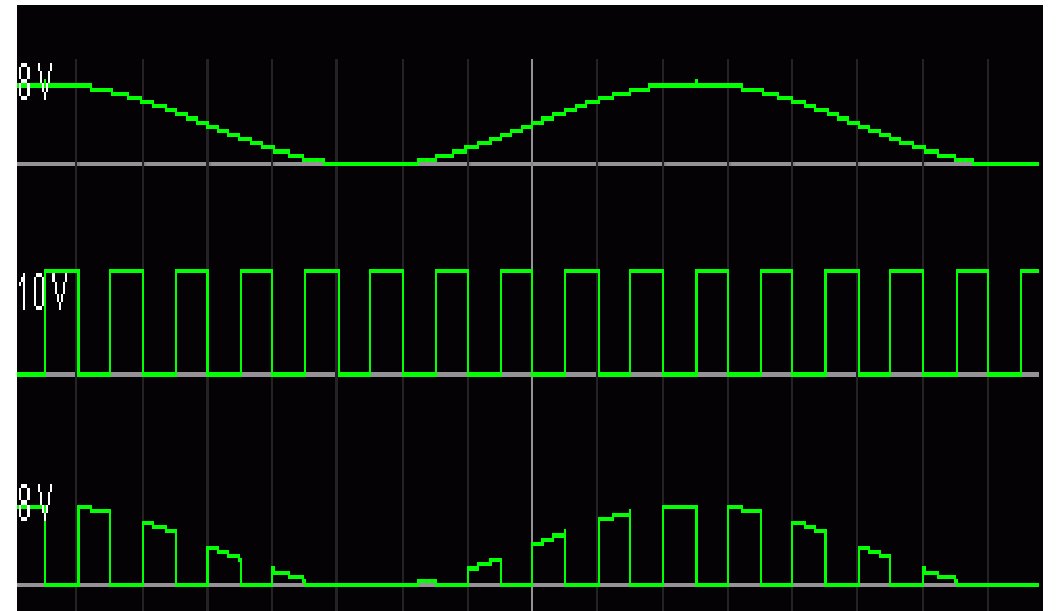
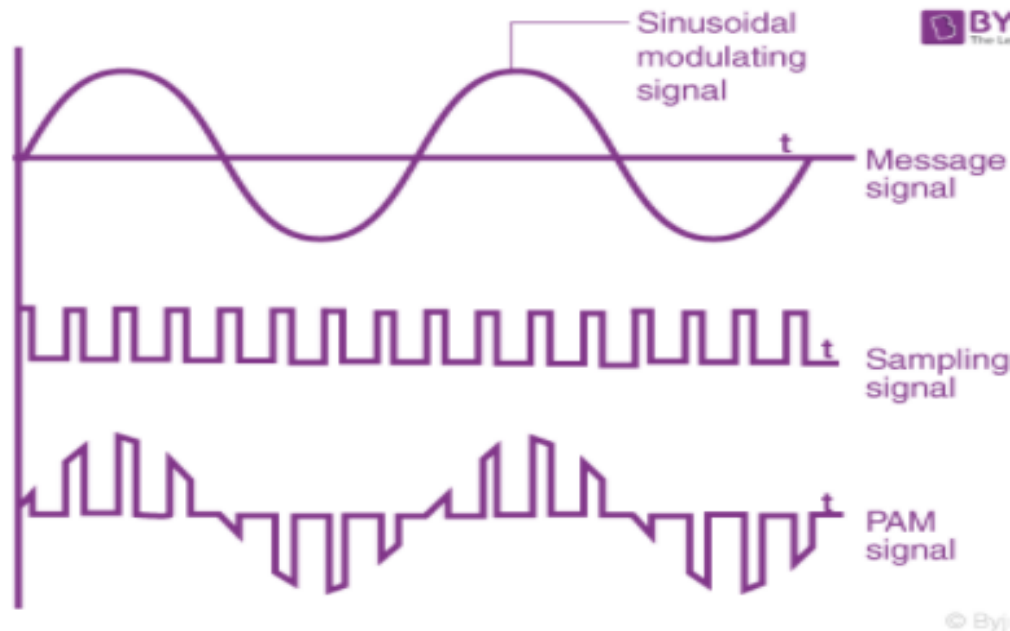
Waveform representation of PAM signal

Electronics Coach

In PAM , the amplitude of the pulses of the carrier pulse train is varied in accordance with the modulating signal

Pulse Amplitude Modulation(PAM)

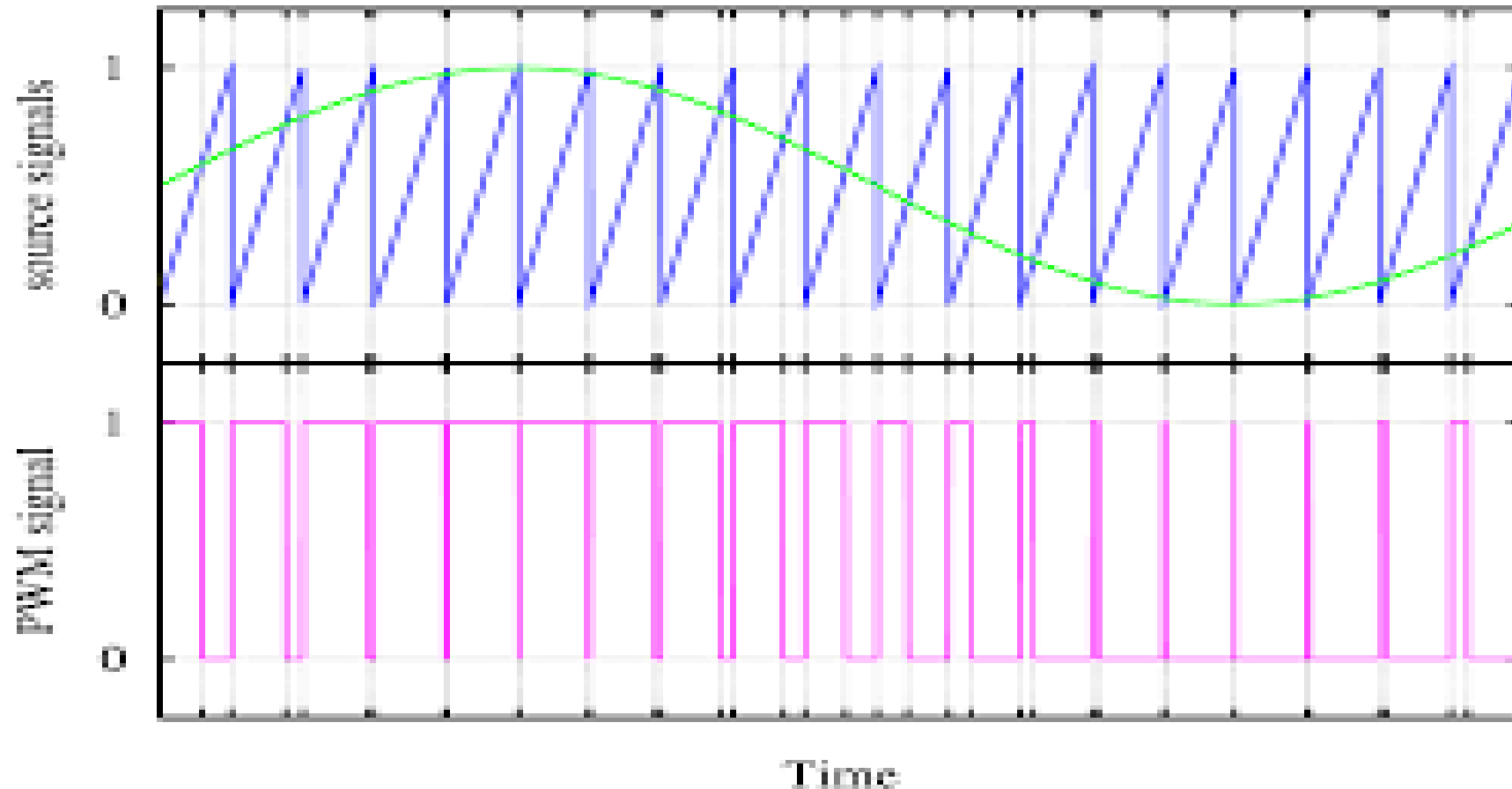
- Simplest form of pulse modulation
- Signal is sampled at regular intervals and each sample is made proportional to the amplitude of the signal at the instant of sampling



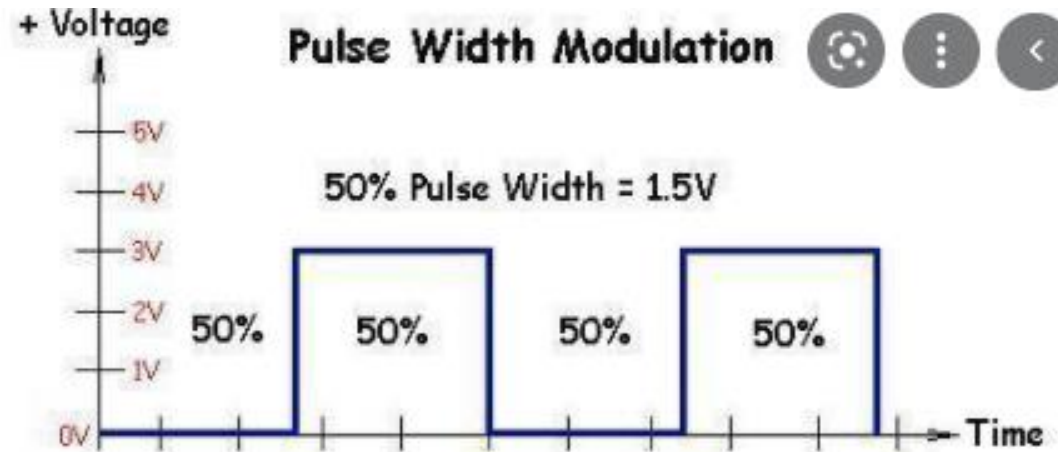
Pulse width or Pulse duration modulation(PWM or PDM)

- ❑ Starting time and amplitude of the pulse are constant but the width or duration of each pulse is made proportional to the instantaneous value of analog signal
- ❑ Disadvantage
- ❑ Pulses are of varying width and hence of varying power content.
- ❑ The transmitter must be powerful to handle the maximum width pulses

Pulse width or Pulse duration modulation(PWM or PDM)



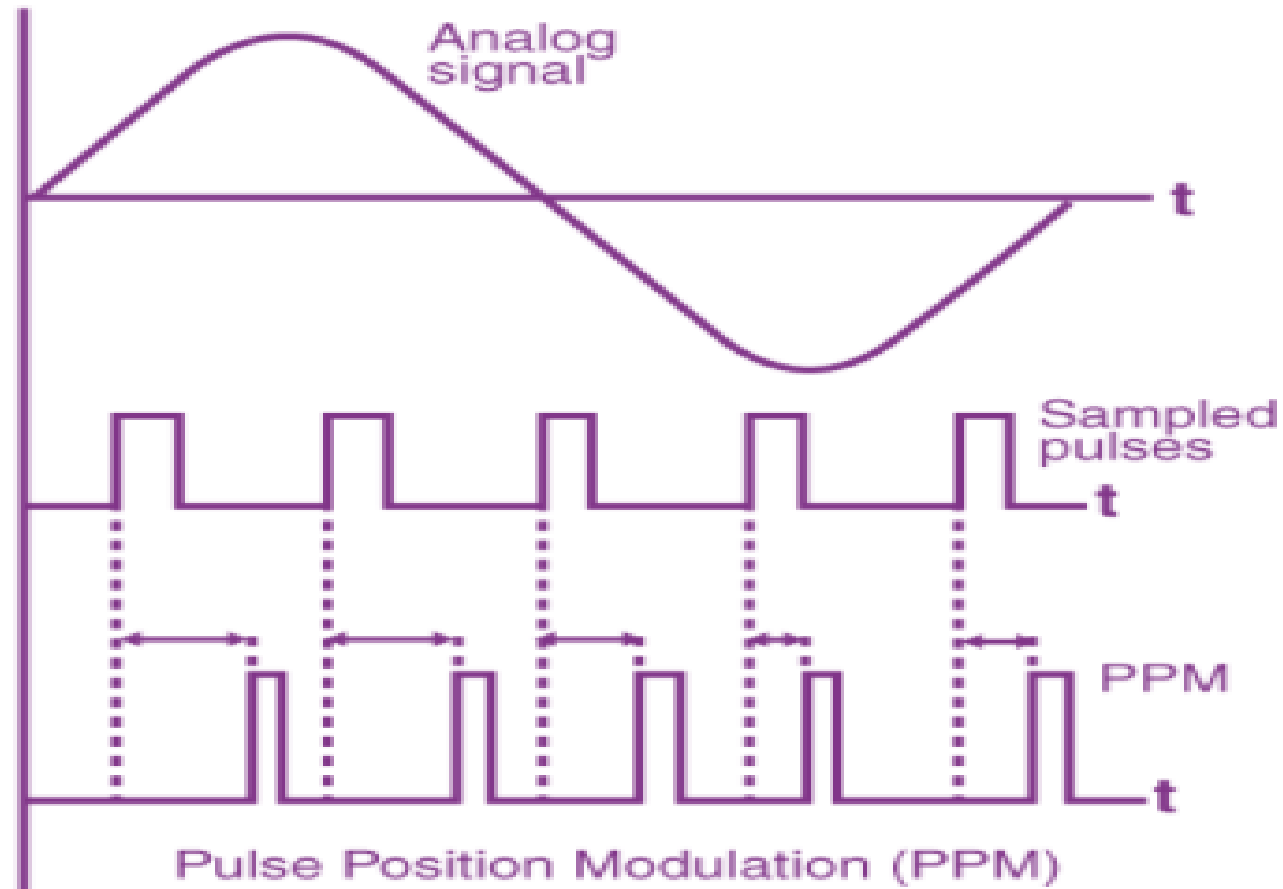
Pulse width or pulse-duration modulation (PWM or PDM)



Pulse Position Modulation (PPM)

Amplitude and width of the pulses are constant but the position of each pulse in relation to the position of the reference pulse is varied according to the instantaneous sampled value of the modulating signal

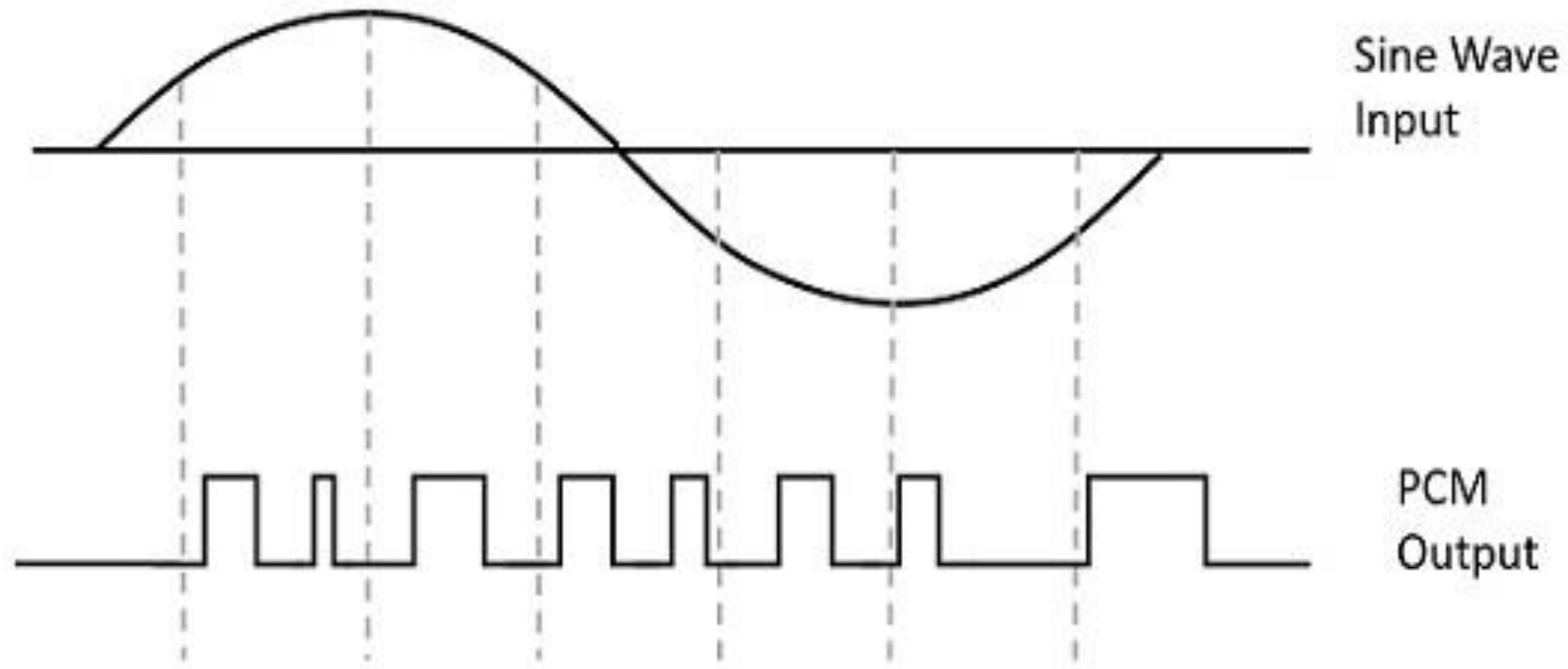
Pulse Position Modulation (PPM)



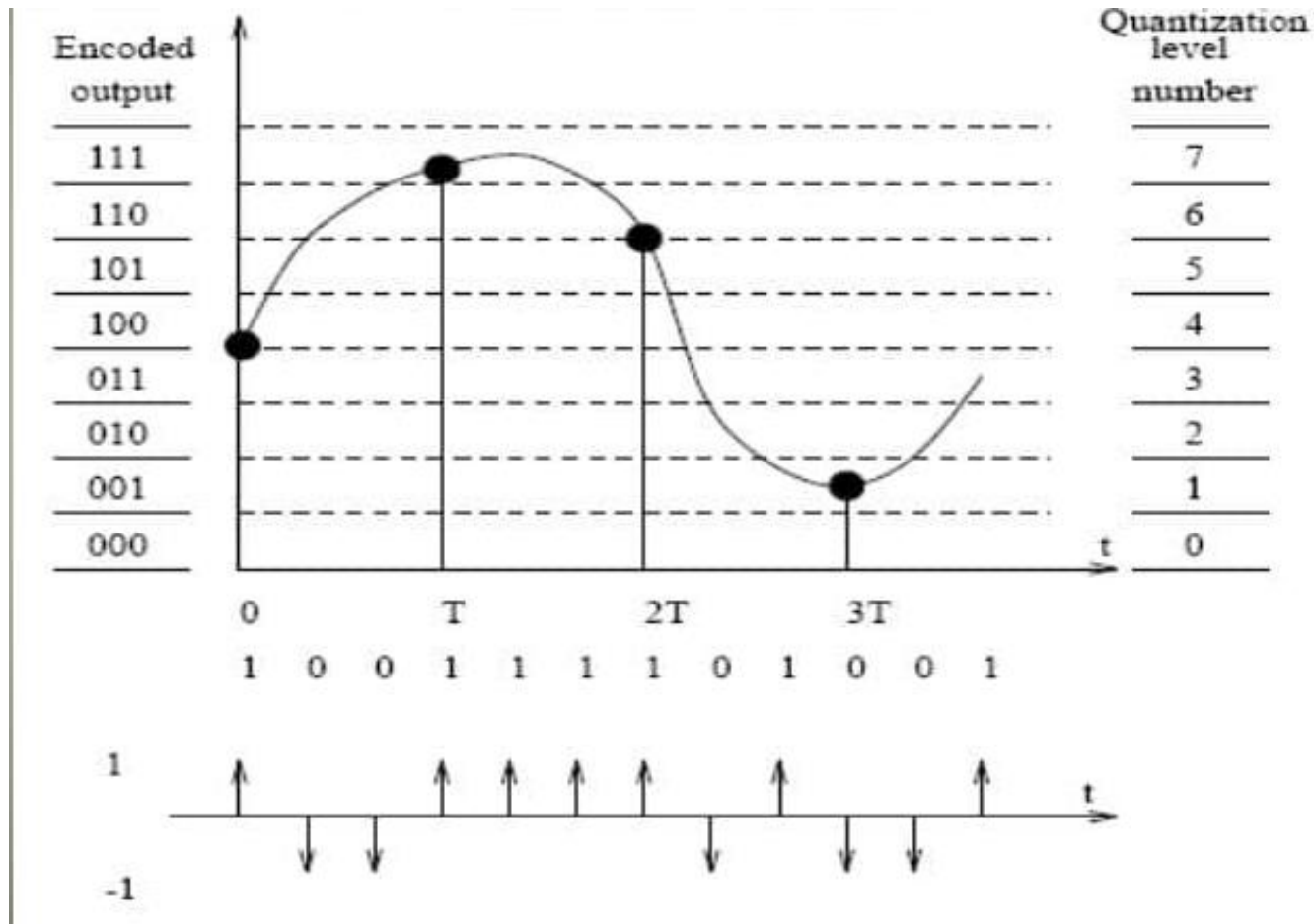
Pulse Code Modulation(PCM)

- ❑ A digital process in which the message is sampled and rounded off to the nearest value of a finite set of allowable values
- ❑ The rounded off values are coded
- ❑ PCM generator produces a series of numbers or digits
- ❑ Each of these digits in binary code represents the amplitude of the signal sample at that instant.

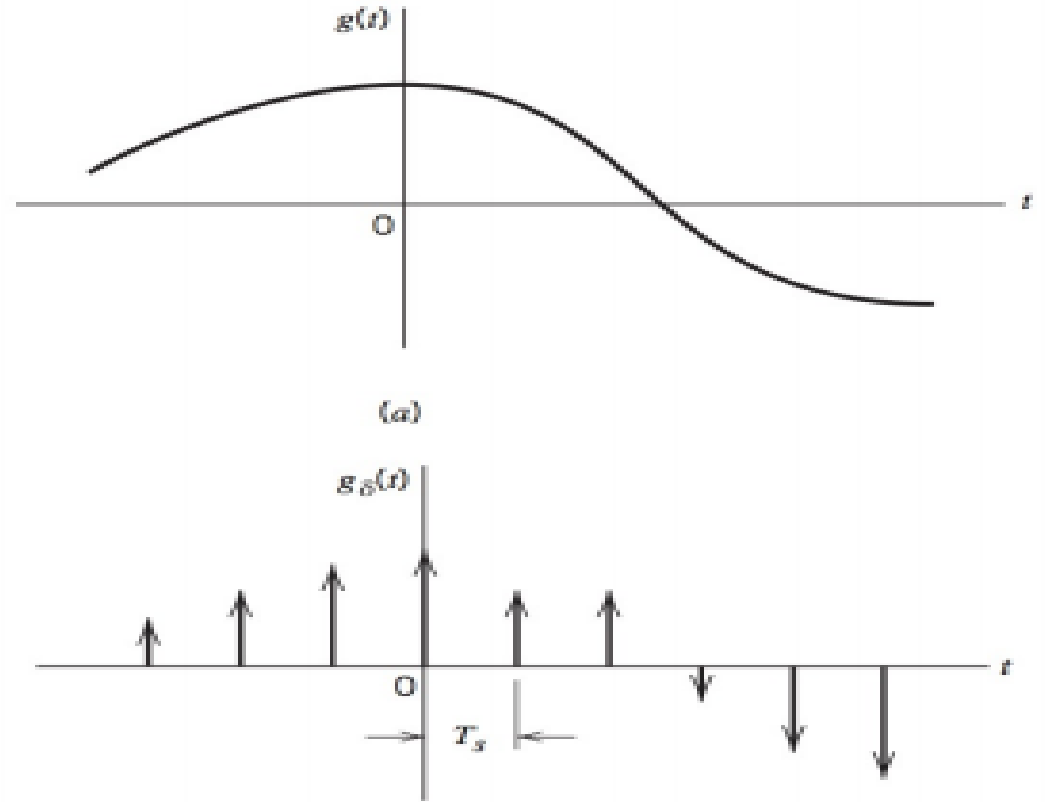
Pulse-code modulation (PCM)



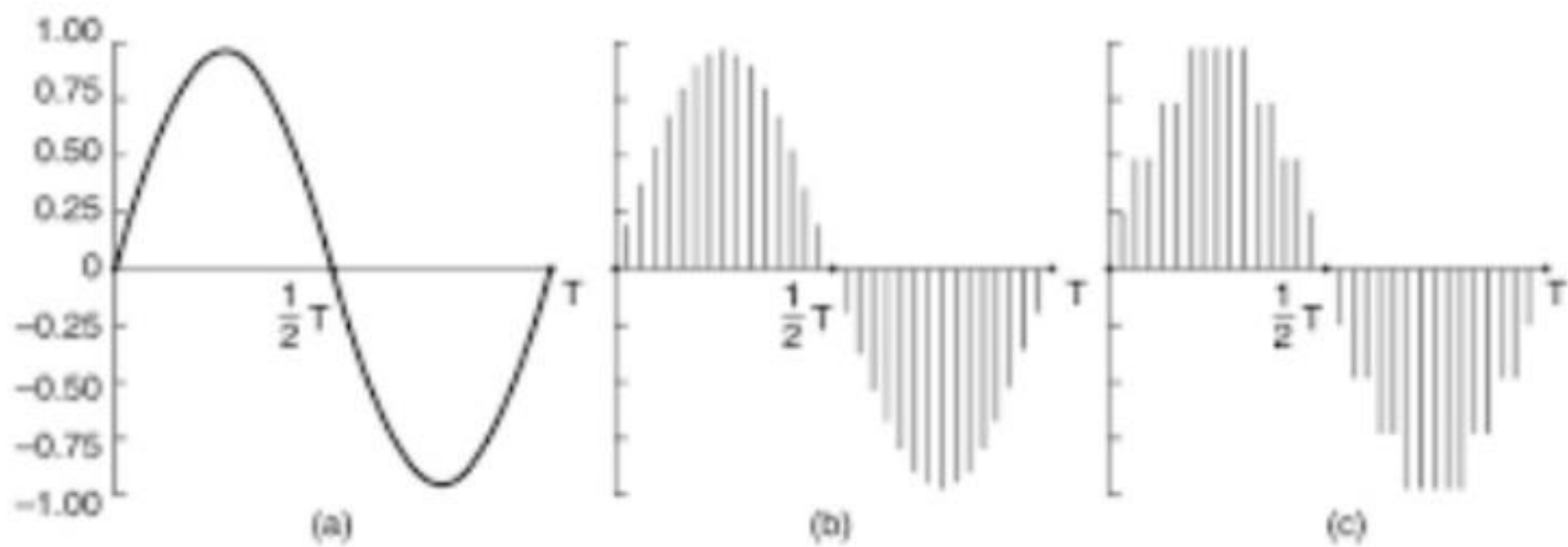
Pulse Code Modulation(PCM)



Sampling



- It is necessary that we choose the sampling rate properly
- Sequence of samples uniquely defines the original analog signal



Sampling Theorem

Sampling theorem :

States that any continuous time signal can be completely represented in its samples and recovered back if the sampling frequency is greater than or equal to twice the highest frequency component of base band signal. That is Sampling frequency, $f_s > 2W$.

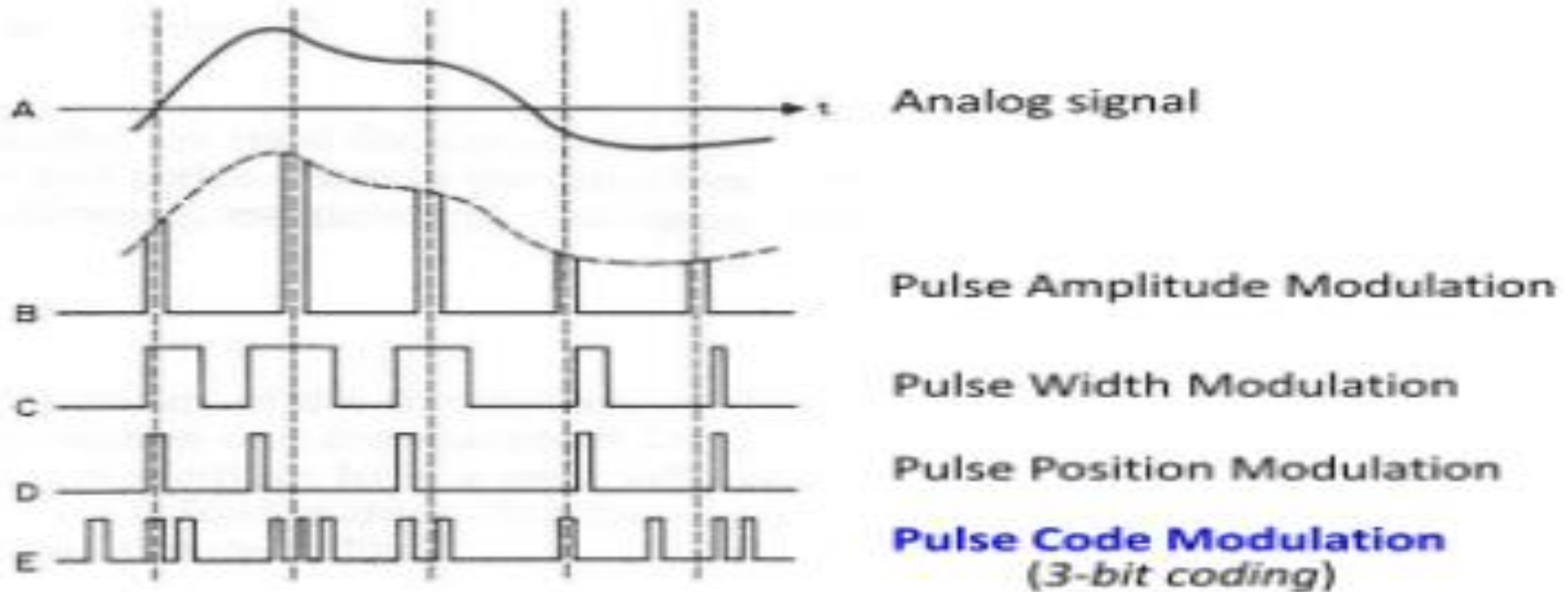
- Where W = Highest frequency in base band continuous time signal.

This condition is also called **Nyquist condition** for sampling process.

Summary: Pulse modulation techniques

Pulse Code Modulation

ES442 – Spring 2016



Radio wave propagation

- Depending on the frequency, a radio wave travels from the transmitting to the receiving antenna in several ways.
- On the basis of the mode of propagation, radio waves can be broadly classified as:
 - (i) ground or surface wave
 - (ii) space or tropospheric wave, and
 - (iii) sky way

Ground wave propagation

- Radio waves are guided by the earth and move along its curved surface from the transmitter to the receiver.
- they are strongly influenced by the electrical properties of the ground.
- useful only at low frequencies as high frequency waves are strongly absorbed by ground.
- Below 500 kHz, ground waves can be used for communication within distances of about 1500 km from the transmitter.
- AM radio broadcast in the medium frequency band cover local areas and take place primarily by the ground wave.
- Ground wave transmission is very reliable whatever the atmospheric conditions be.

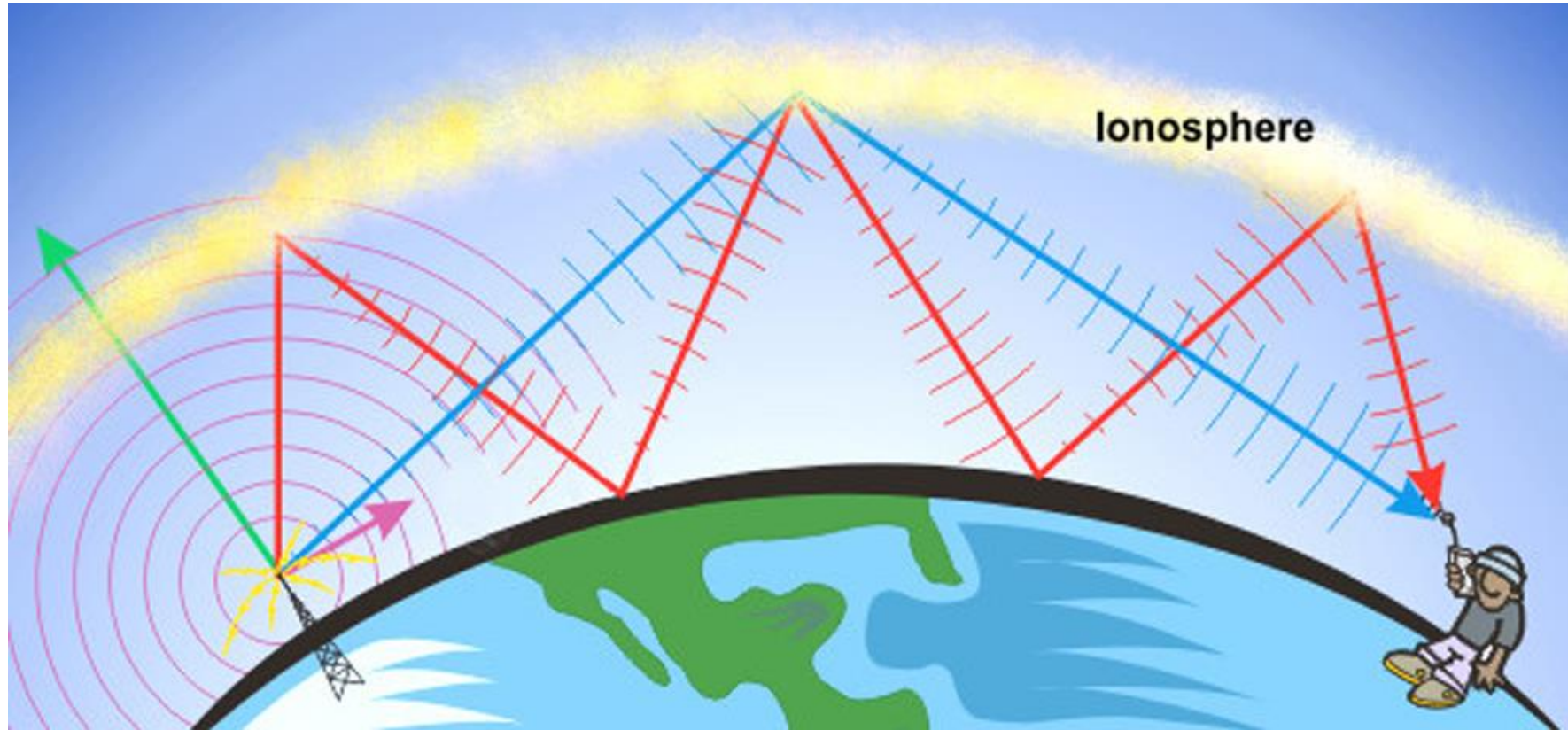
Space or tropospheric wave propagation

- Radio wave transmitted from an antenna, travels in a straight line directly reaches the receiving antenna
- In space wave or line of sight propagation, radio waves move in the earth's troposphere within about 15 km over the surface of the earth.
- The space wave is made up of two components:
 - (a) a *direct or line-of-sight* wave from the transmitting to the receiving antenna and
 - (b) the *ground-reflected* wave traversing from the transmitting antenna to ground and reflected to the receiving antenna
- Television frequencies in the range 100-220 MHz are transmitted through this mode.

Sky wave propagation

- radio waves transmitted from the transmitting antenna reach the receiving antenna after reflection from the ionosphere, i.e. the ionized layers lying in the earth's upper atmosphere.
- Short wave transmission around the globe is possible through sky wave via successive reflections at the ionosphere and the earth's surface.
- *ionosphere* - The ionized *region* of the earth's upper atmosphere extending from about 40 km to the height of a few earth radii above the earth.
- The ionosphere is made up of electrons, and positive and negative ions in the background of neutral particles of the atmosphere.
- *The propagation of radio wave through the ionosphere* is affected by the electrons and ions in the ionosphere. The effect of the electrons on the propagation is much greater than that of the ions since the electronic mass is much less than the ionic mass.

Ionosphere

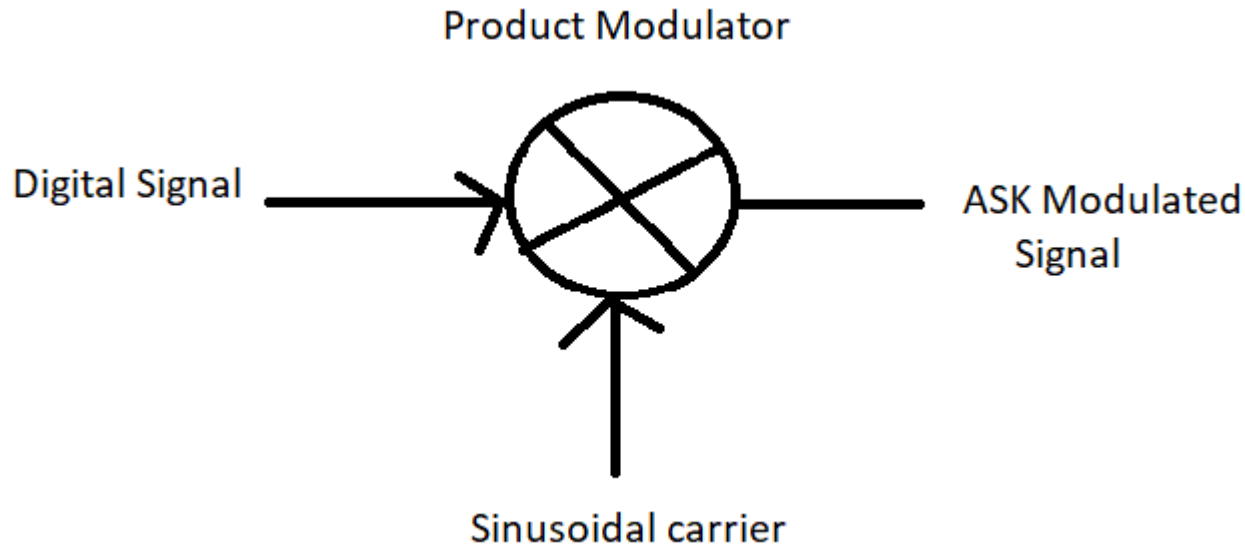


Digital Modulation Schemes

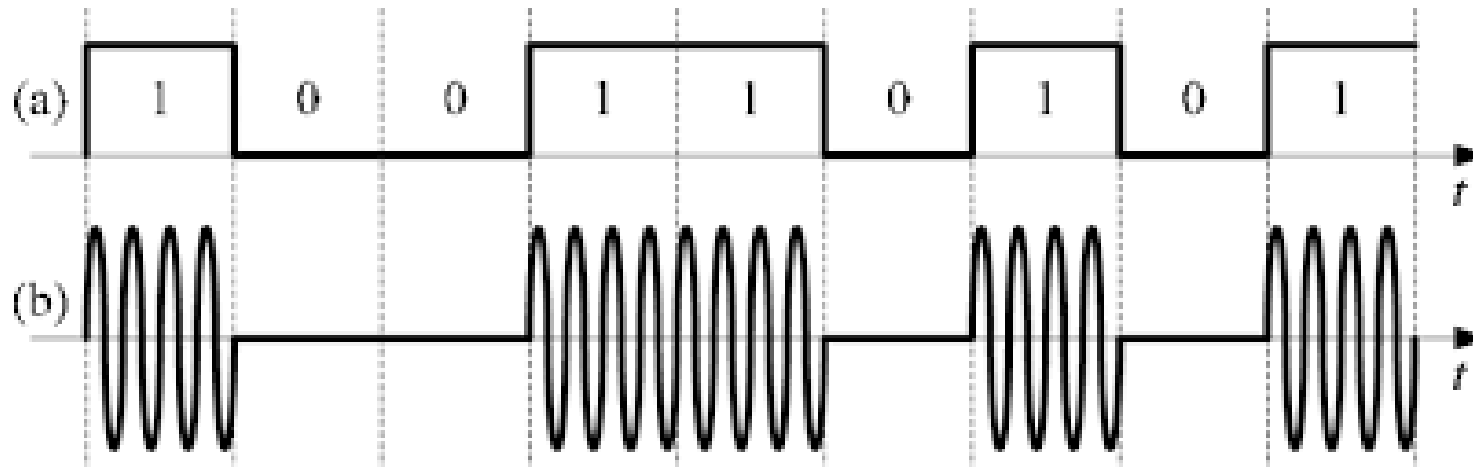
- In digital communications, the modulating signal consists of binary data or M-ary version of it.
- When it is required to transmit digital signals, the amplitude, frequency or phase of the sinusoidal carrier is varied in accordance with the incoming digital data.
- Digital modulation schemes are classified as
 - Amplitude Shift Keying (ASK)
 - Frequency Shift Keying (FSK)
 - Phase Shift Keying (PSK)

Amplitude Shift Keying(ASK)

- Represents digital data as variations in the amplitude of a carrier wave
- ASK signal may be generated by simply applying the incoming binary data and the sinusoidal carrier to the inputs of a product modulator.

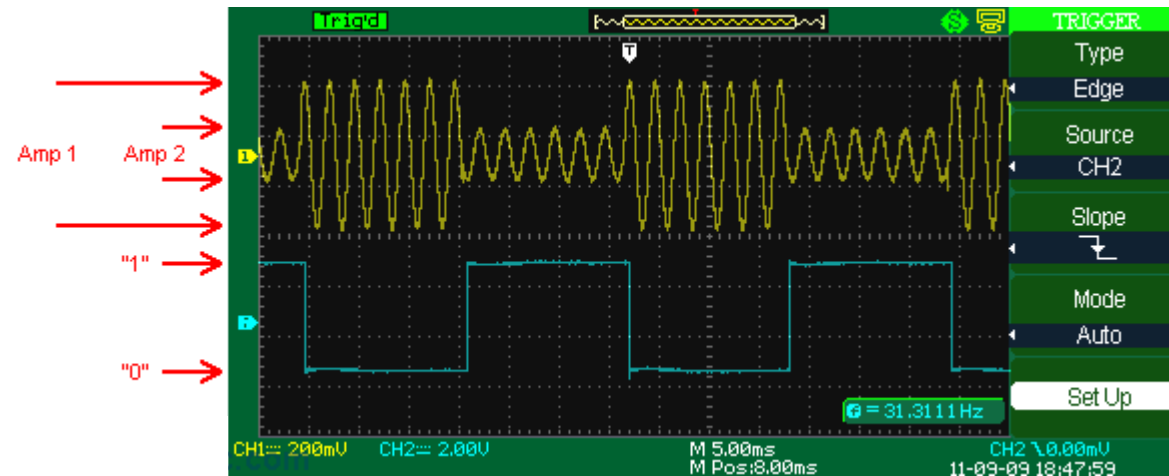


Amplitude Shift Keying(ASK)



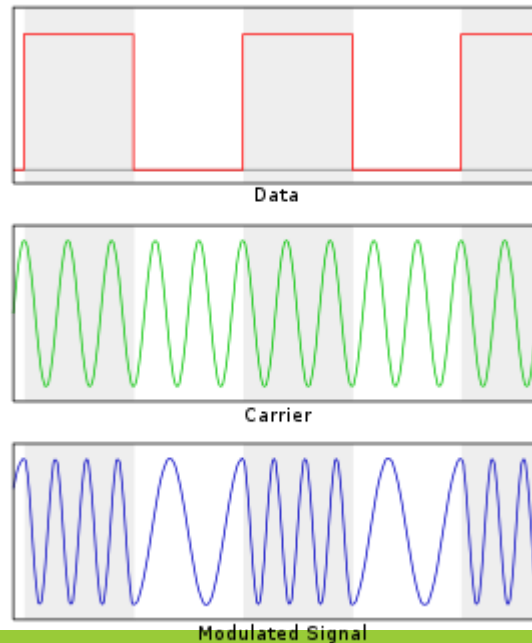
ASK Modulation

Amplitude Shift Keying(ASK)

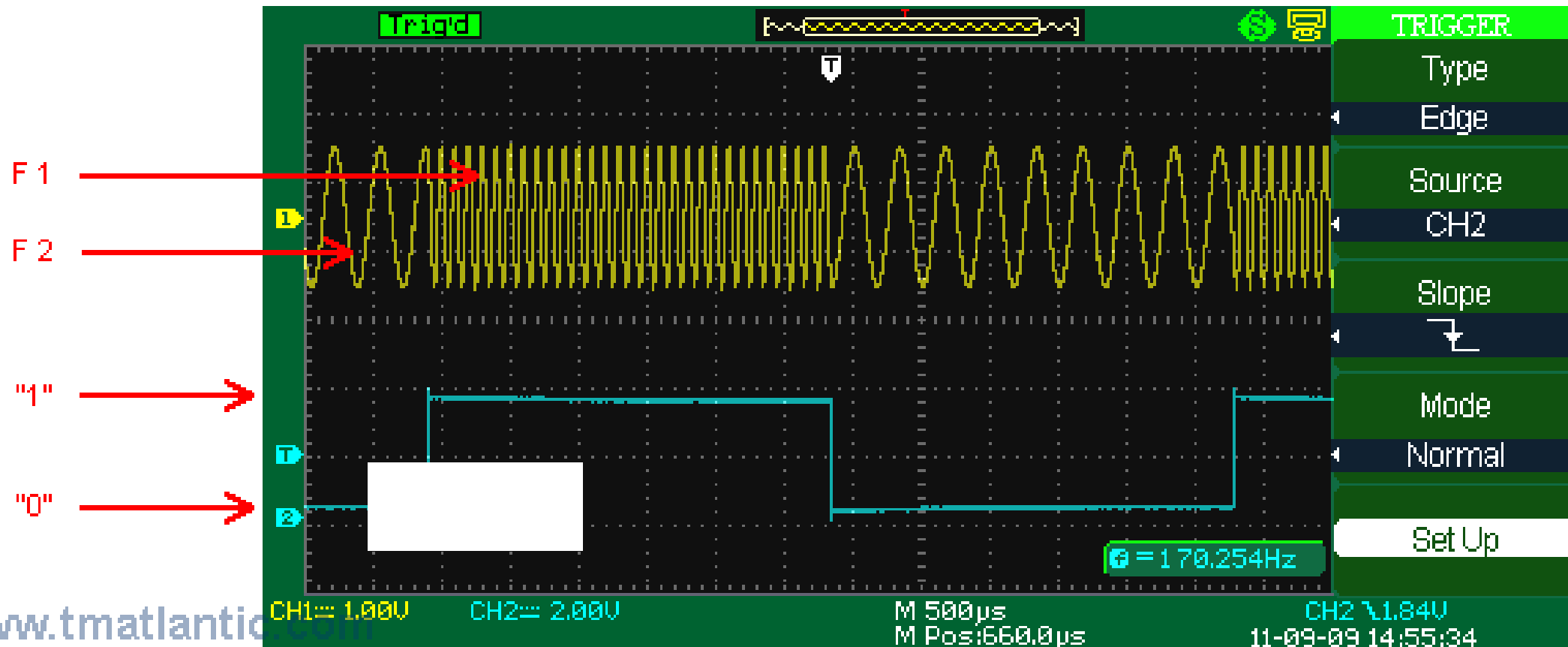


Frequency Shift Keying(FSK)

- Digital information is transmitted through discrete frequency changes of a carrier signal
- The simplest FSK is **binary FSK (BFSK)**.
- BFSK uses a pair of discrete frequencies to transmit binary (0s and 1s) information



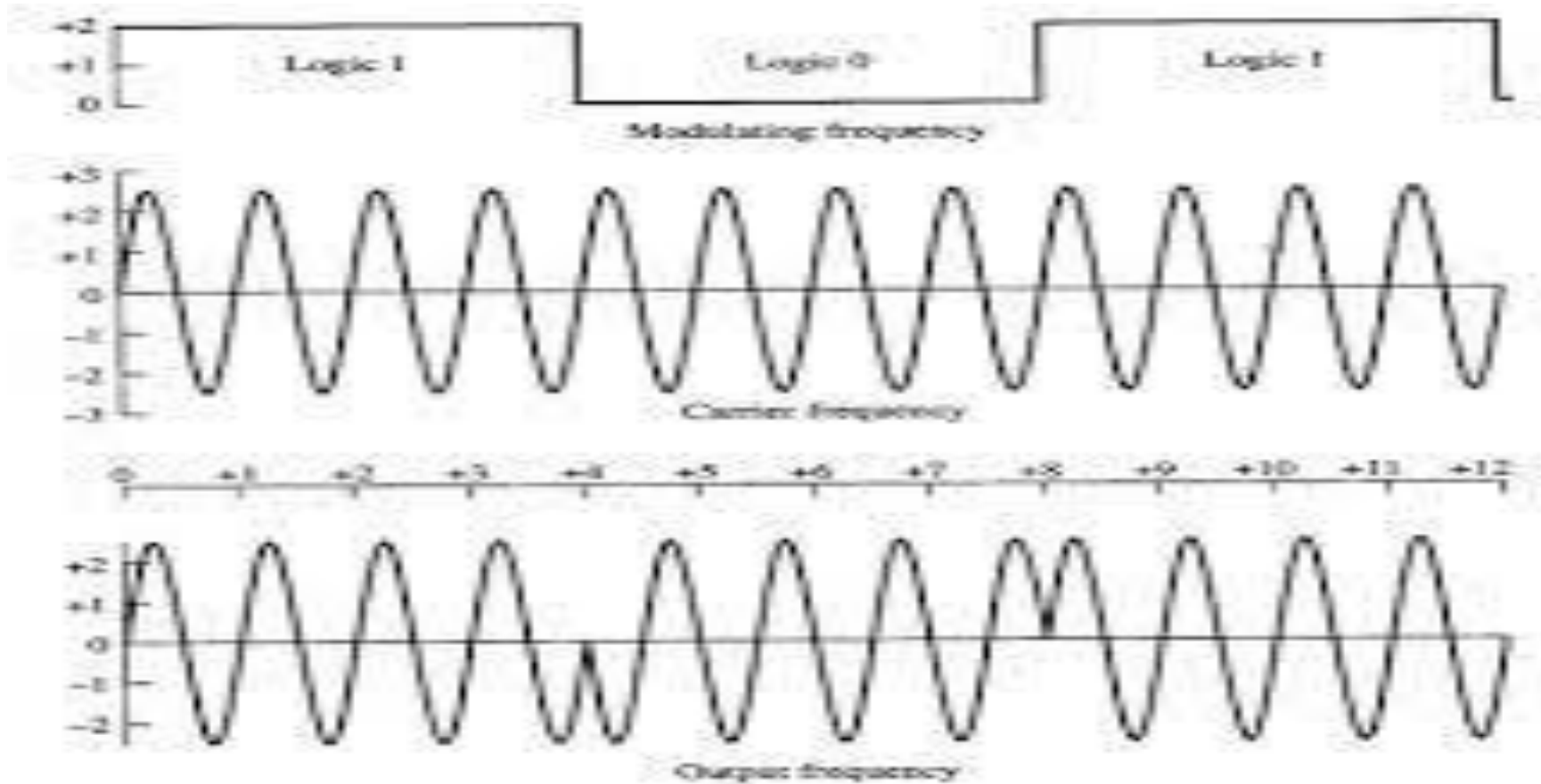
Frequency Shift Keying(FSK)



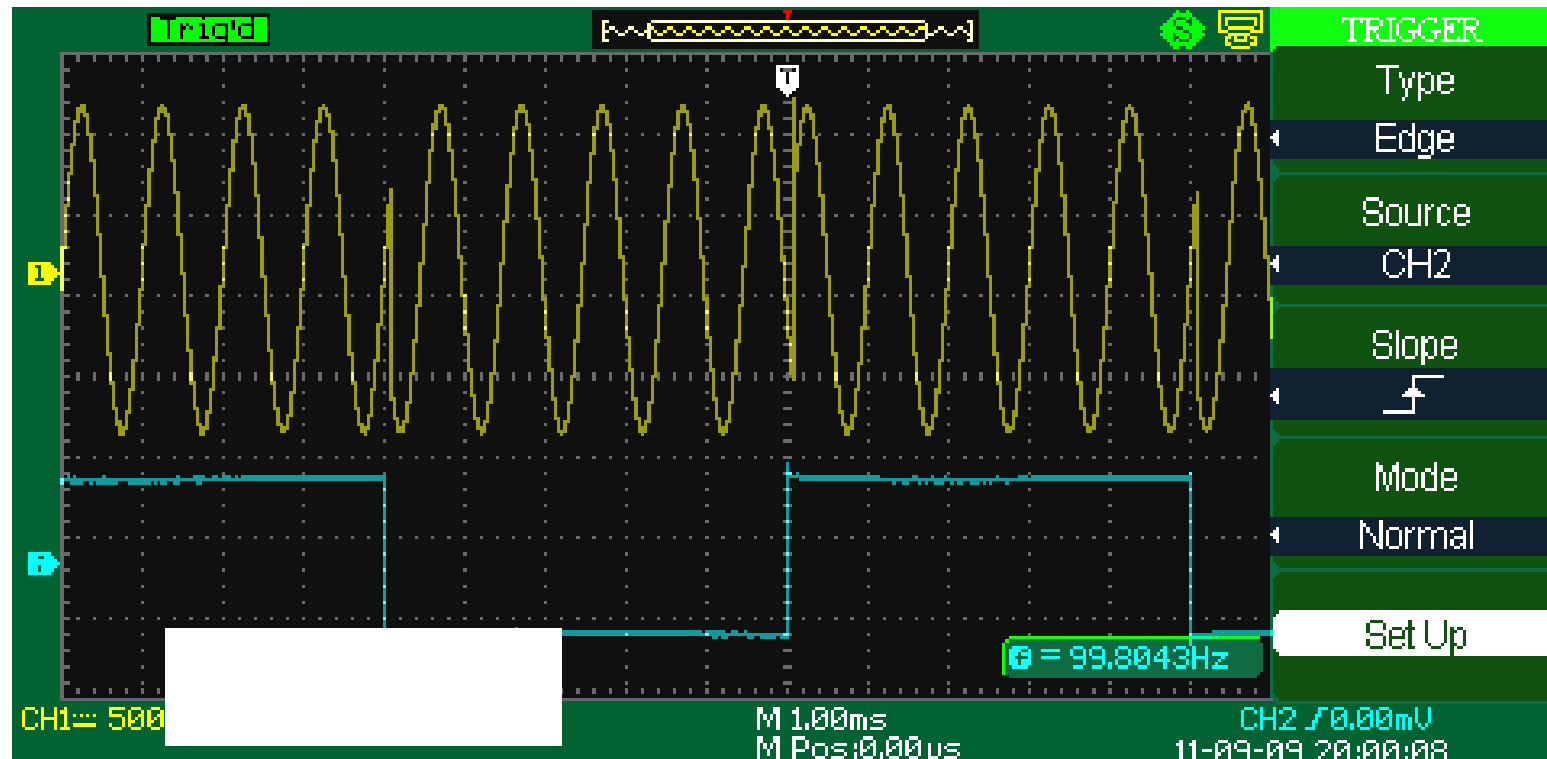
Phase Shift Keying(PSK)

- Conveys data by changing (modulating) the phase of constant frequency carrier
- Each symbol(pattern of bits) is represented by a particular phase.
- BPSK (Binary PSK), the simplest form of PSK, uses phases 0^0 and 180^0
- It is widely used for wireless LANs, RFID and Bluetooth communication

Phase Shift Keying(PSK)



Phase Shift Keying(PSK)



PSK and FSK more preferred over ASK?

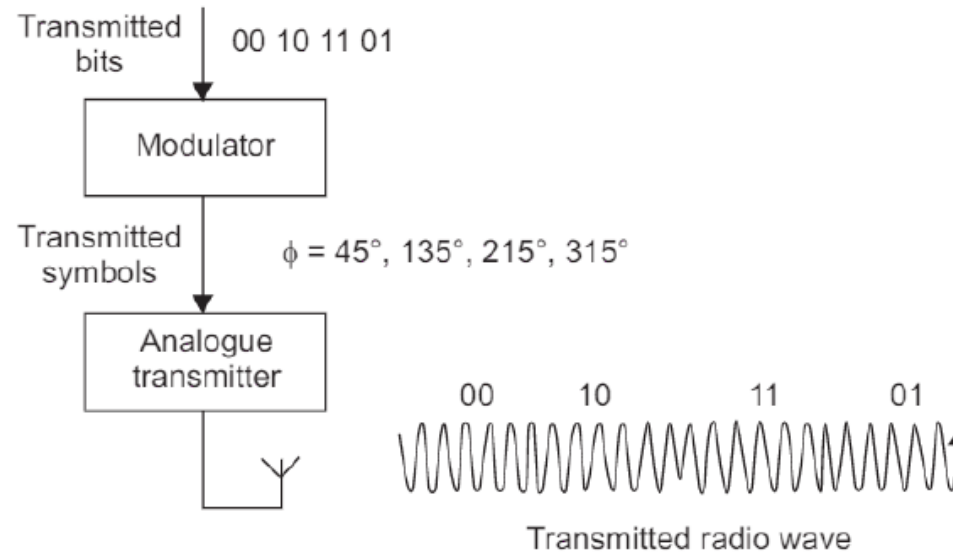
- Because of the constant amplitude of FSK or PSK, the effect of noise and interference is minimum whereas it is more pronounced in ASK

Link for ASK, PSK and FSK video

- <https://www.youtube.com/watch?v=C8eebS5MhuU>

Radio signal transmission

Important components of wireless transmission system



Note : The modulation scheme used here is QPSK

Architecture of wireless communication transmitter

Radio signal transmission

The transmitter accepts a stream of bits from the application software and encodes these bits onto a radio wave, known as a carrier, by adjusting parameters of the wave such as its amplitude or phase.

- Transmitter processes the incoming bits in two stages
 - 1. Modulator** – accepts the incoming bits and computes symbols that represent the amplitude and phase of the outgoing wave.

Modulation scheme used –QPSK – the modulator takes 2 bits at a time and transmits them using radio waves having 4 different states with phases 45° , 135° , 225° and 315°

- 2. Analogue transmitter** – Generates the radio wave

Modulation scheme used - QPSK

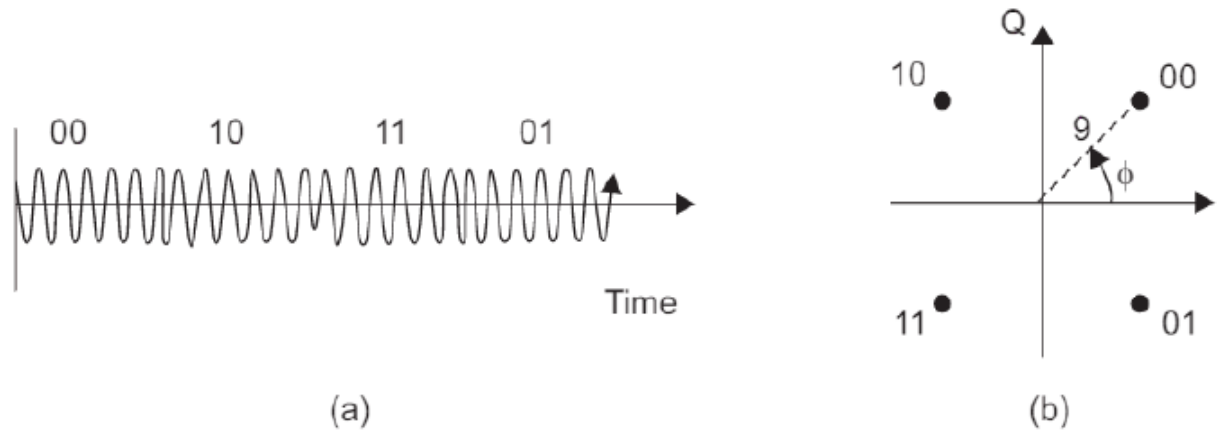


Fig (a) QPSK Waveform

(b) QPSK Constellation Diagram

- The distance of each state from the origin represents the amplitude of the transmitted wave while angle measured anti-clockwise from x-axis represents the phase

Modulation schemes used in LTE

LTE ((Long Term Evolution(LTE) - standard for wireless broadband communication for mobile devices marketed as 4G)

- Binary Phase Shift Keying - BPSK
- Quadrature Phase Shift Keying - QPSK
- 16 Quadrature Amplitude Modulation - 16 QAM
- 64 Quadrature Amplitude Modulation - 64 QAM

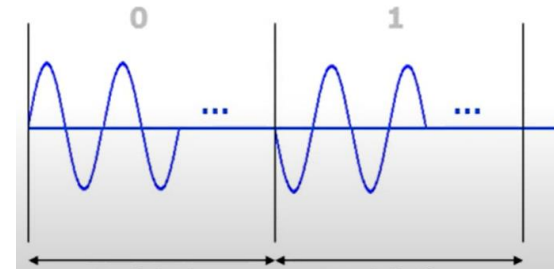
Binary Phase Shift Keying (BPSK)

Binary Phase Shift Keying (BPSK) - sends bits one at a time, using 2 states that can be interpreted as phases 0 and 180 or as amplitudes +1 and -1.

One bit per symbol



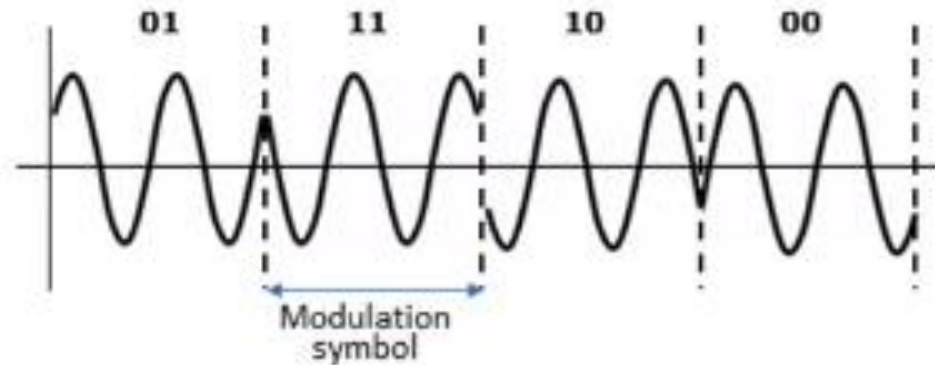
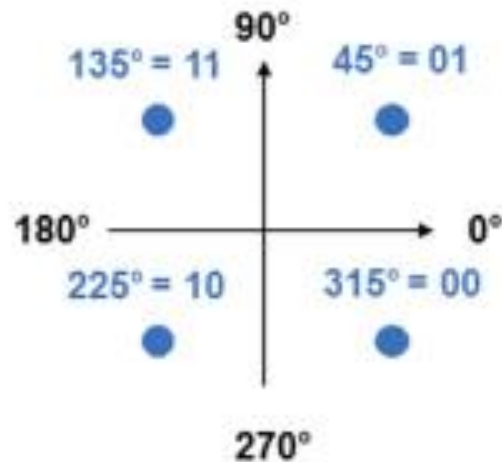
Constellation diagram of BPSK



Note : LTE uses this scheme for limited number of control streams but does not use it for normal data transmissions

Quadrature Phase Shift Keying (QPSK)

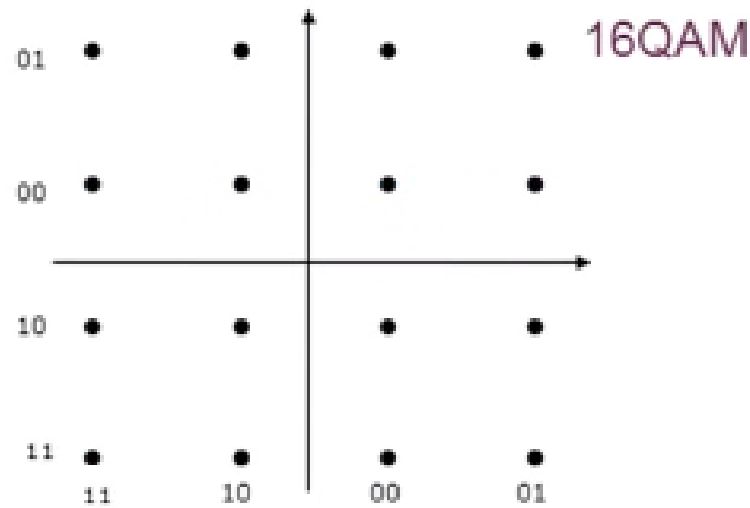
Quadrature Phase Shift Keying (QPSK) – sends 2 bits per symbol at a time, using 4 states that can be interpreted as phases 45, 135, 215 and 315 which correspond to bit combinations of 00, 10, 11 and 01 respectively.



Constellation diagram of QPSK

16 Quadrature Amplitude Modulation (16 - QAM)

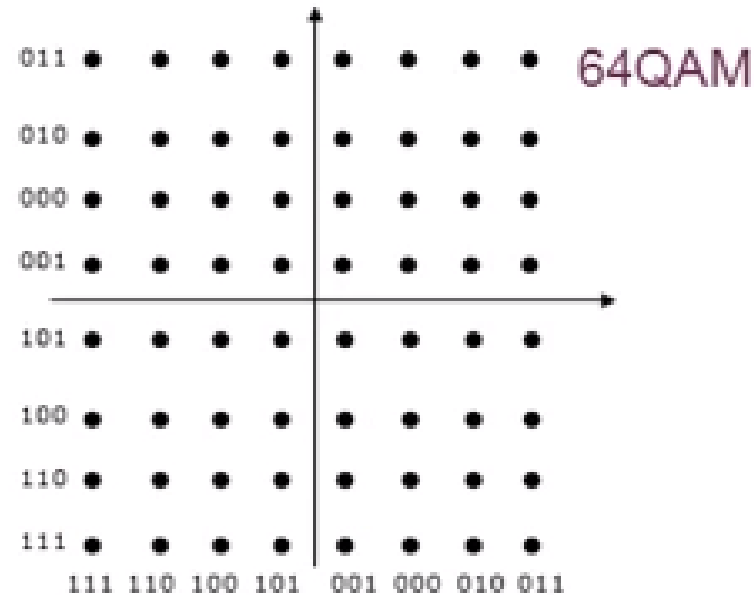
16 Quadrature Amplitude Modulation (16 - QAM) – Sends bits 4 at a time, using 16 states that have different amplitudes and phases



Constellation diagram of 16 - QAM

64 Quadrature Amplitude Modulation (64-QAM)

- **64 Quadrature Amplitude Modulation (64-QAM)** – Sends 6 bits per symbol at a time using 64 different states.
- Data rate is six times greater than that of BPSK.



YouTube video for constellation diagram:
<https://www.youtube.com/watch?v=kfJeL4LQ43s>

Constellation diagram of 16 - QAM

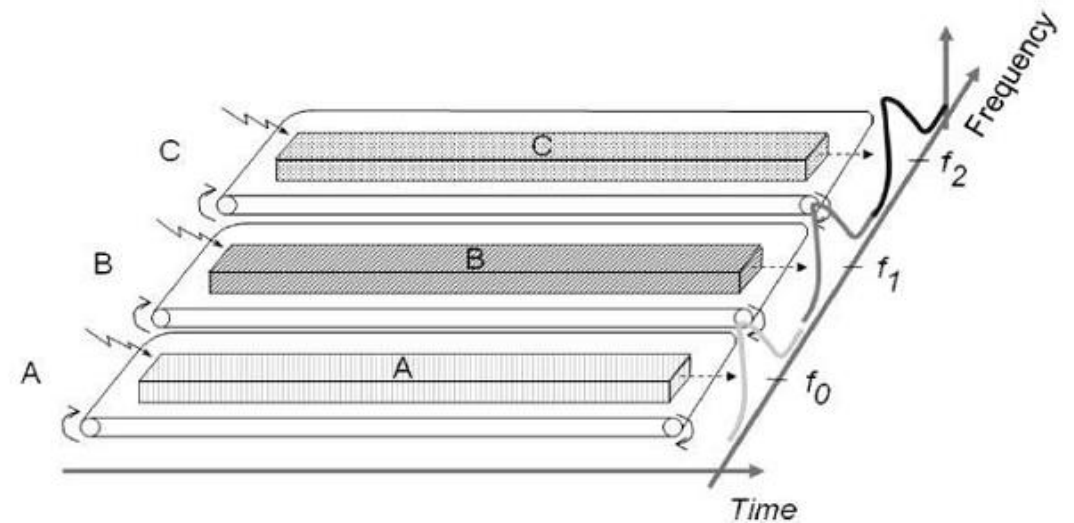
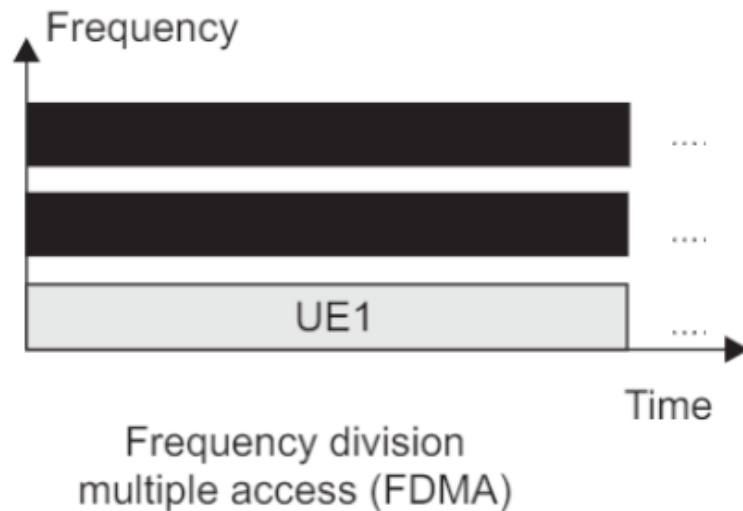
Multiple Access Techniques

- In a wireless environment where most of the mobile users are trying to transmit their signal it is difficult to provide interference free transmission.
 - Solution is multiple access techniques.
-
- In cellular network, the base station has to transmit to many different mobiles at once by sharing the resources of air interface using a technique known as multiple access.
 - Multiple access techniques
 - Frequency Division Multiple Access (FDMA)
 - Time Division Multiple Access (TDMA)
 - Code Division Multiple Access (CDMA)

Frequency Division Multiple Access (FDMA)

Each mobile receives signal on its own carrier frequency, which it distinguishes using analogue filter.

The carriers are separated by unused guard rings to minimize the interference between the carriers



FDMA

- Frequency Division Multiple Access
- channel spectrum divided into number of non overlapping frequency bands with some guard-band.
- each station assigned fixed frequency band
- Each Channel Occupies the same frequency bands all the time.
- Receiver should tune to the same frequency band to receive the signal.

Advantages & Disadvantages of FDMA

- Advantages:

- All the users can transmit their information all the time without waiting for their turn.
- No Dynamic Co-ordination is required

- Disadvantages:

- Inflexible and inefficient if channel load is dynamic and uneven.
- Adjacent channel interference is present in spite of providing guard-bands.

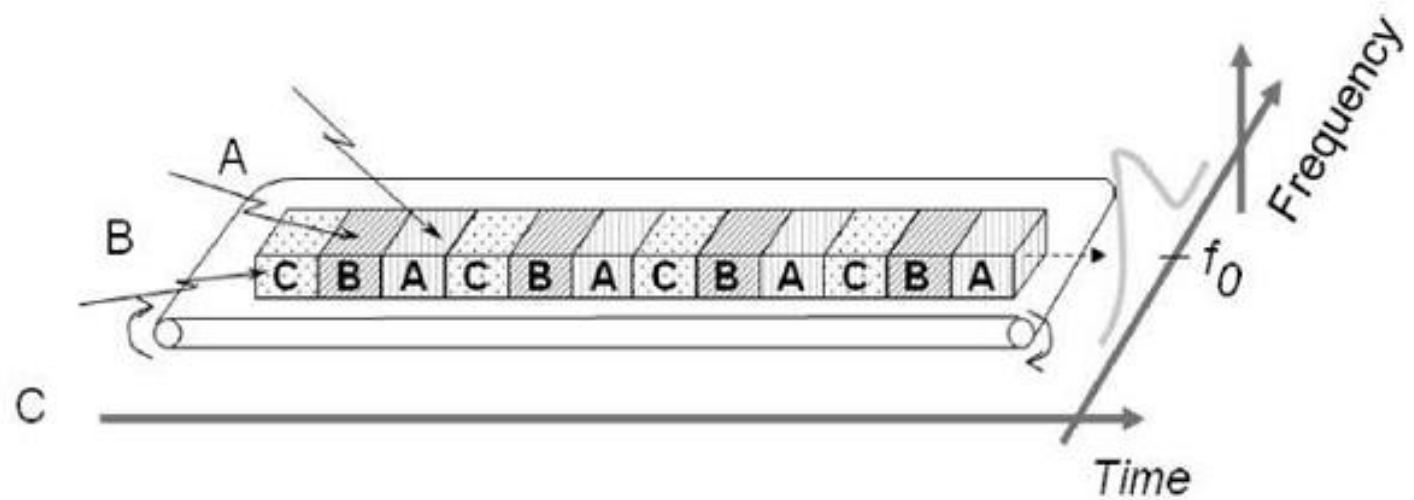
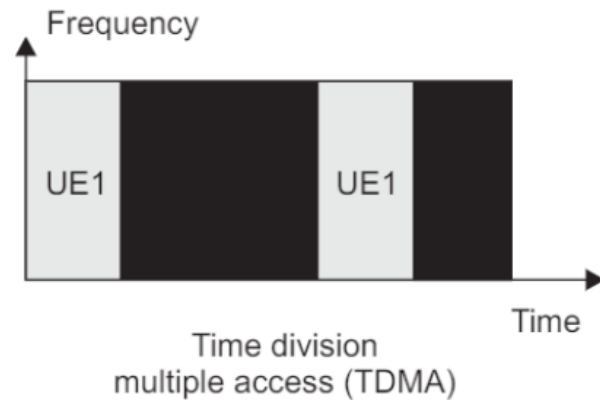
Applications of FDMA

Applications:

- Each channel gets a band (range) of frequencies used in traditional radio, TV etc.
- FDMA is also present in various wireless communication standards such as AMPS, CT2, DECT etc.

TIME Division Multiple Access (TDMA)

Mobiles receive signals on the same carrier but at different times.



TDMA

- Time Division Multiple Access
- Each user can access to channel in "rounds"
- each user gets fixed length slot (length = pkt transmission time) in each round
- Each user can transmit the signal using same frequency but at different time.
- Entire bandwidth is available to the user but only for small duration of time i.e. time slot duration.

Advantage:

- Can assign more time to sender with heavier Load.
 - No need for precise narrow band filters.
 - Extended battery life.
 - Higher capacity.
-

Disadvantage:

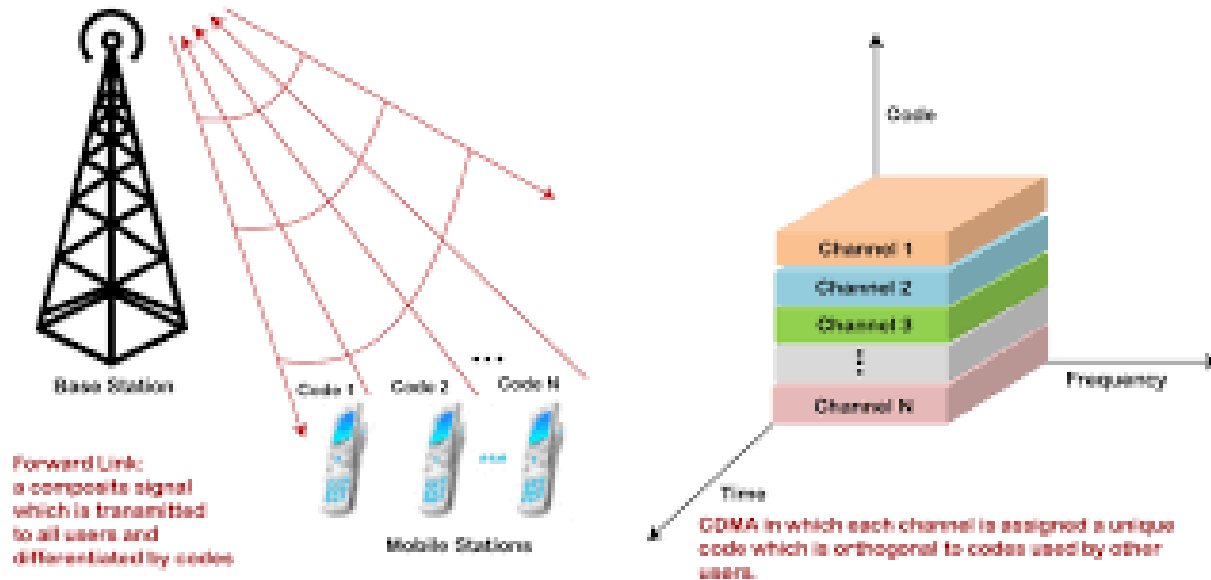
- Require timing Synchronization.

Applications of TDMA

- TDMA is used in the digital 2G cellular systems such as Global System for Mobile Communications (GSM),

Code Division Multiple Access (CDMA)

- Technique used by third generation communication systems
- Mobiles receive signals on the same carrier frequency and at the same time.
- The signals are labelled by the use of codes, which allows a mobile to separate its own signal from the others.



CDMA

- Code Division Multiple Access
- All users use the same frequency but each user has its own “chipping sequence” i.e. code to encode the data.

Encoding:

- Encoded signal = (chipping sequence) X (original data)

Decoding:

- Inner product of encoded signal and chipping sequence.

CDMA

- Allows multiple users to “co-exists” and transmits simultaneously with minimal interference if and only if the codes are orthogonal to each other.
- Each station transmits over the entire spectrum all the time but are not garbled.
- Multiple simultaneous transmission are separated using coding theory.

Advantages:

- bandwidth efficient - code spaces is huge
 - Almost impossible to jam
 - Larger capacity and low power consumption
-

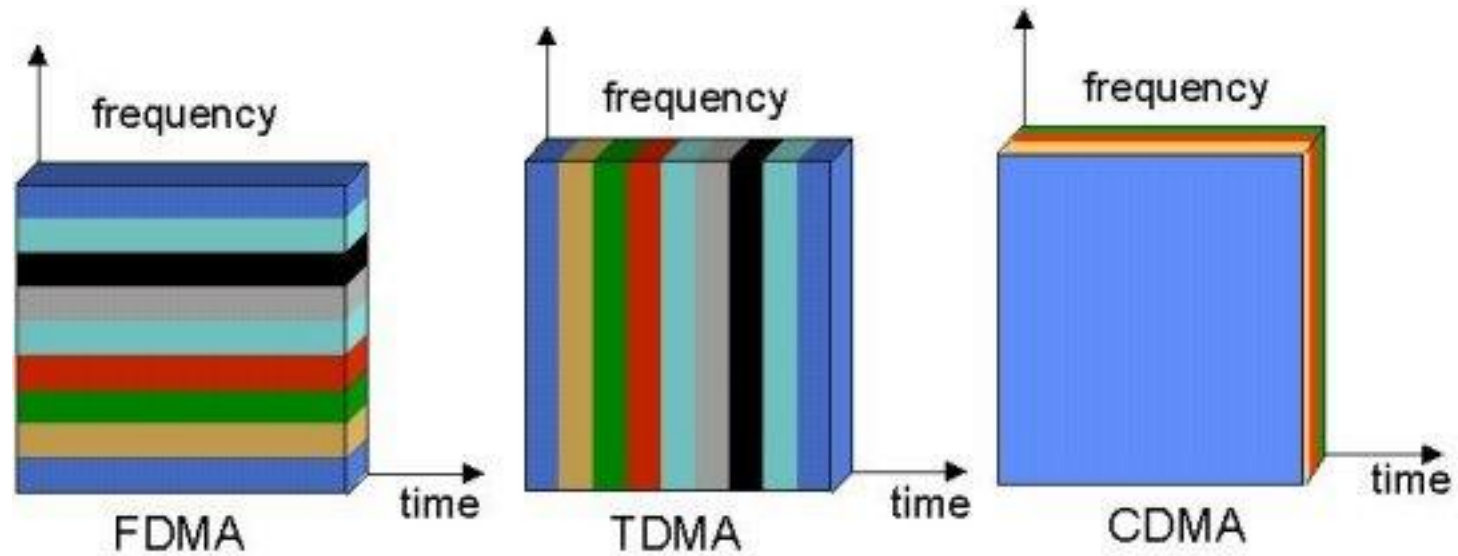
Disadvantages:

- more complex signal regeneration
- Implemented using Spread Spectrum.

Applications of CDMA

- Mainly used in wireless broadcast channels such as cellular, satellite etc.
- CDMA is used in IS-95, WCDMA, cdma2000 etc.
- CDMA has merged with the GSM technology to give a high-speed 4G or LTE internet services.

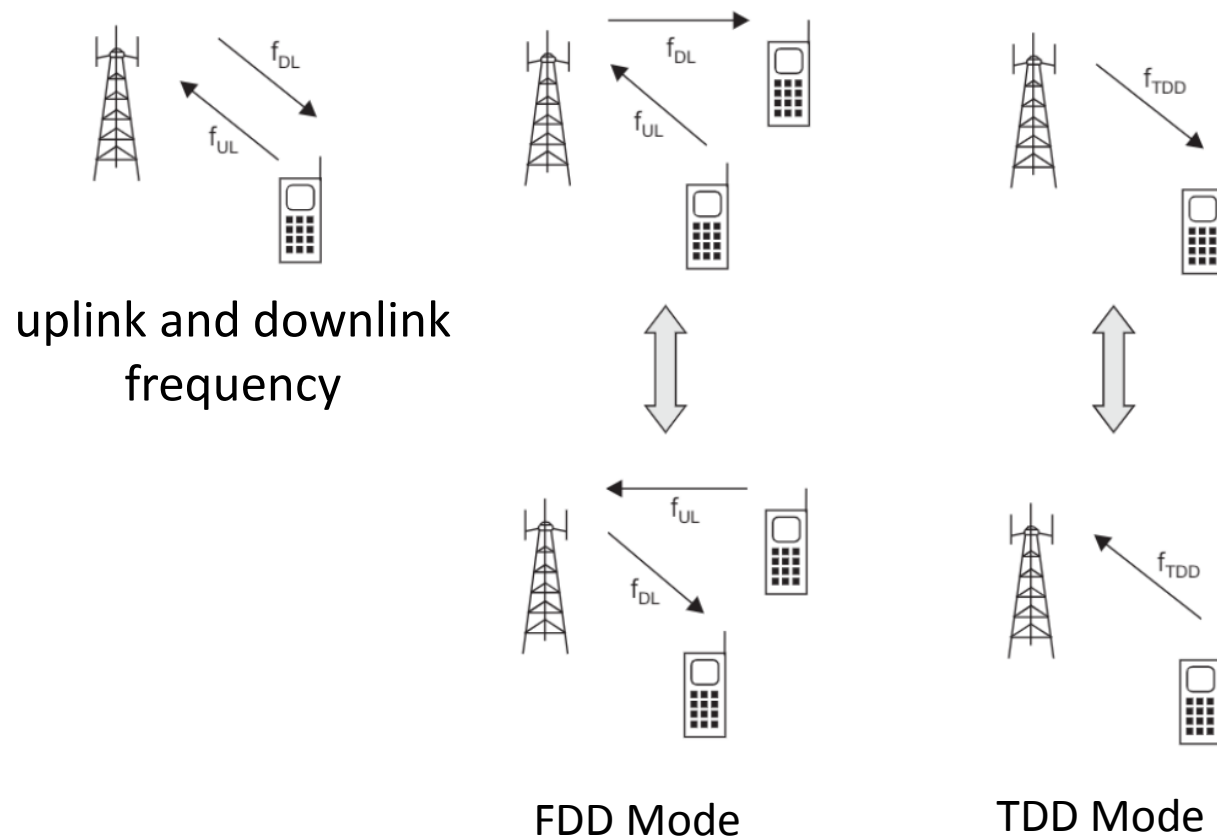
FDMA, TDMA & CDMA



FDD and TDD Modes

- By using the various multiple access techniques, a base station can distinguish the transmissions to and from the individual mobiles in the cell.
- To distinguish the mobiles' transmissions from those of the base stations, a mobile communication system can operate in the transmission modes.
- Frequency Division Duplex (FDD)** -the base station and mobile transmit and receive at the same time, but using different carrier frequencies.
- Time Division Duplex (TDD)** - the base station and mobile transmit and receive on the same carrier frequency but at different times.

Operation of FDD and TDD modes

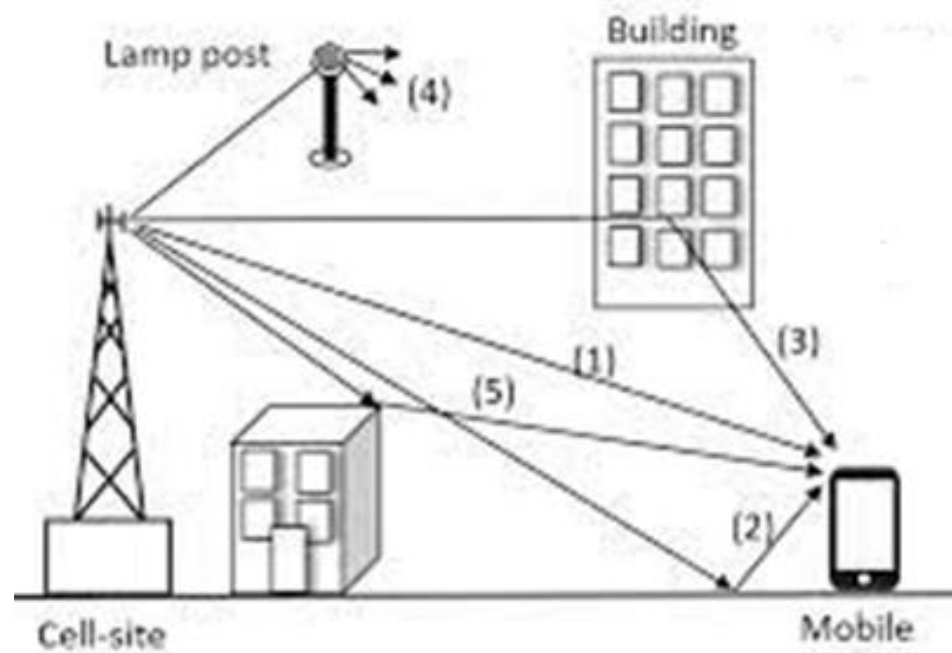


FDD and TDD Modes

- In FDD mode,
 - the bandwidths of the uplink and downlink are fixed and are usually the same
 - suitable for voice communications, in which the uplink and downlink data rates are very similar.
- In TDD mode,
 - the system can adjust how much time is allocated to the uplink and downlink
 - suitable for applications such as web browsing, in which the downlink data rate can be much greater than the rate on the uplink.

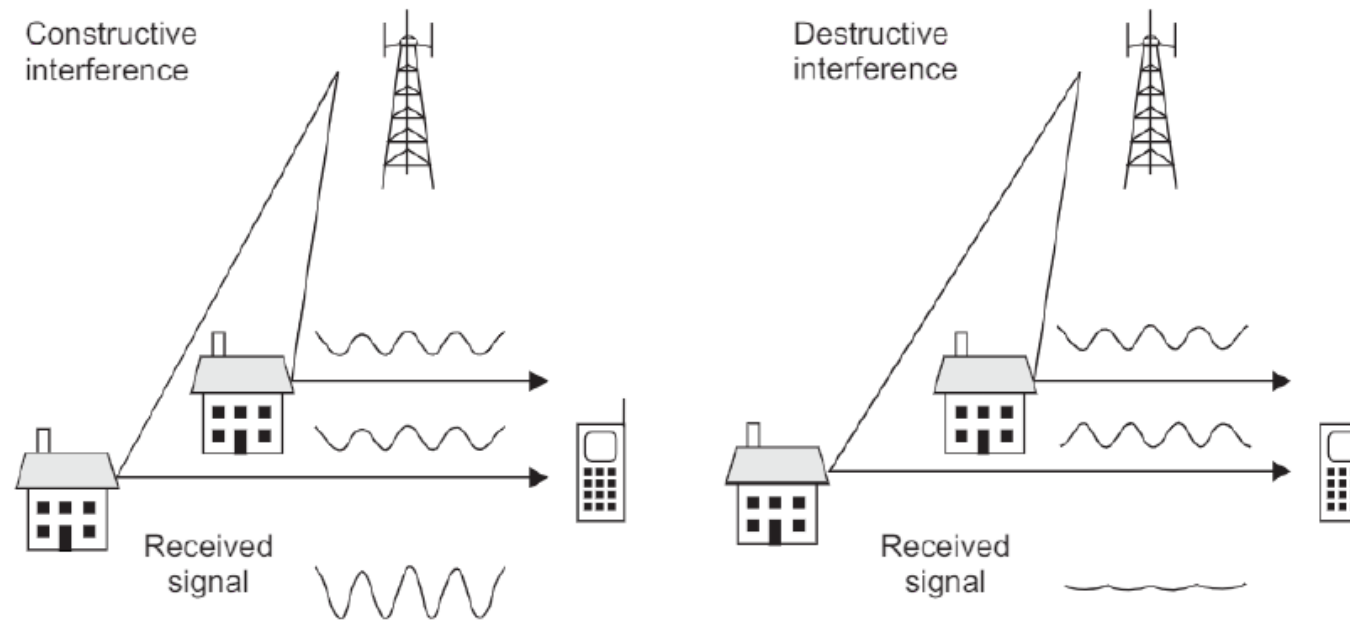
Multipath and Fading

- As a result of reflections, rays can take several different paths from the transmitter to the receiver. This phenomenon is known as **multipath**.



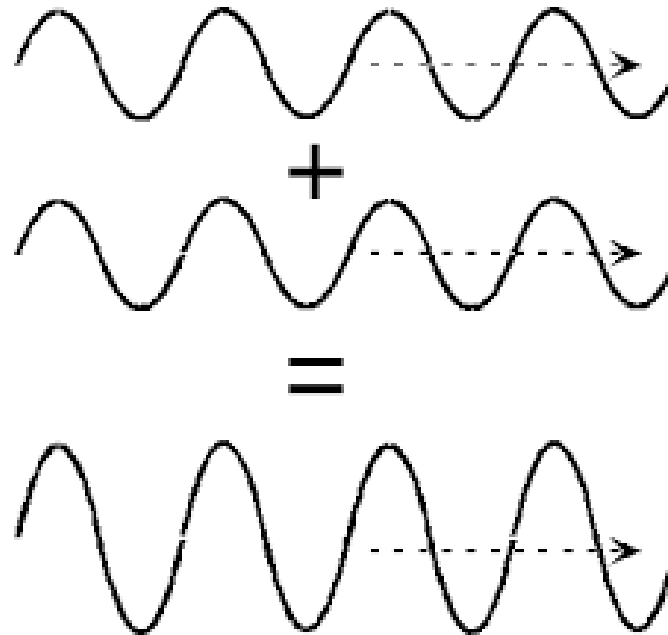
Interference

- At the receiver, the incoming rays can add together in different ways, which are shown in figure below.



Constructive Interference

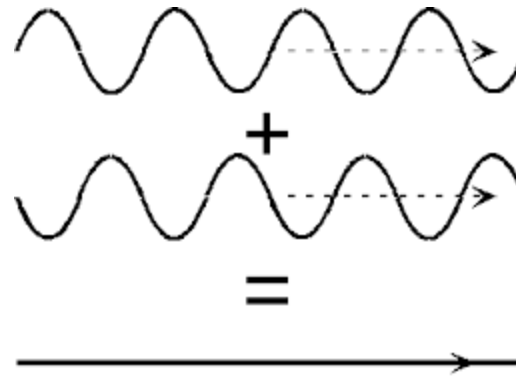
- If the peaks of the incoming rays coincide then they reinforce each other, a situation known as ***constructive interference***.



Constructive Interference

Destructive Interference

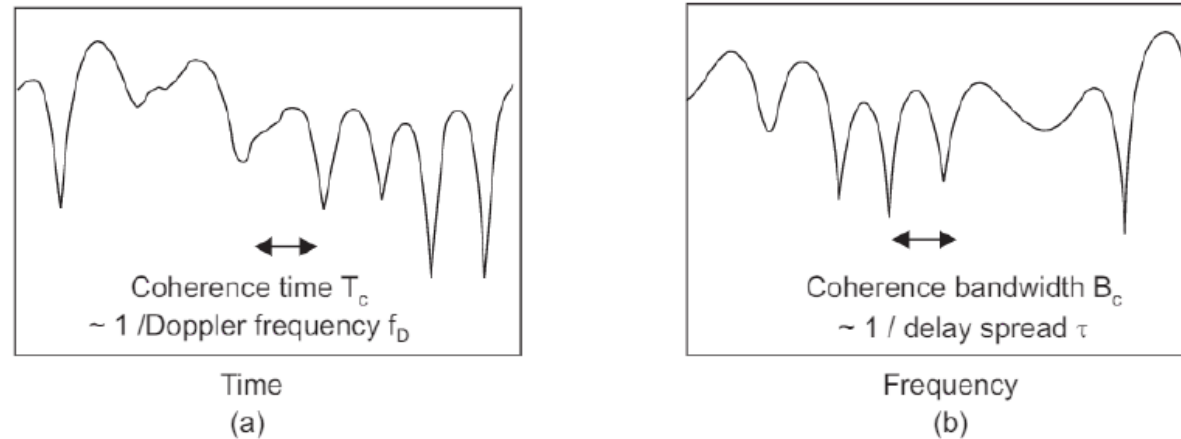
- peaks of one ray coincide with the troughs of another, the result is ***destructive interference***, in which the rays cancel.



- Destructive interference can make the received signal power drop to a very low level, a situation known as ***fading***.

Fading as a function of time and frequency

- If the mobile moves from one place to another, then the ray geometry changes, so the interference pattern changes between constructive and destructive. Fading is therefore a function of time, as shown in Fig.



Fading as a function of (a) time and (b) frequency

Coherence time and Doppler Frequency

- The amplitude and phase of the received signal vary over a timescale called the *coherence time*, T_c which can be estimated as follows:

$$T_c = \frac{1}{f_D}$$

Here f_D is the mobile's Doppler frequency:

$$f_D = \frac{v}{c} f_c$$

- where f_c is the carrier frequency, v is the speed of the mobile and c is the speed of light ($3 \times 10^8 \text{ ms}^{-1}$)

Coherence Bandwidth

- If the carrier frequency changes, then the wavelength of the radio signal changes. This also makes the interference pattern change between constructive and destructive
- The amplitude and phase of the received signal vary over a frequency scale called the *coherence bandwidth*, B_c which can be estimated as follows:

$$B_c = \frac{1}{\tau}$$

Here, τ is the delay spread of the radio channel

Delay spread

- Delay Spread is the difference between the arrival times of the earliest and latest rays. It can be calculated as follows:

$$\Gamma = \frac{\Delta L}{c}$$

where ΔL is the difference between the path lengths of the longest and shortest rays.

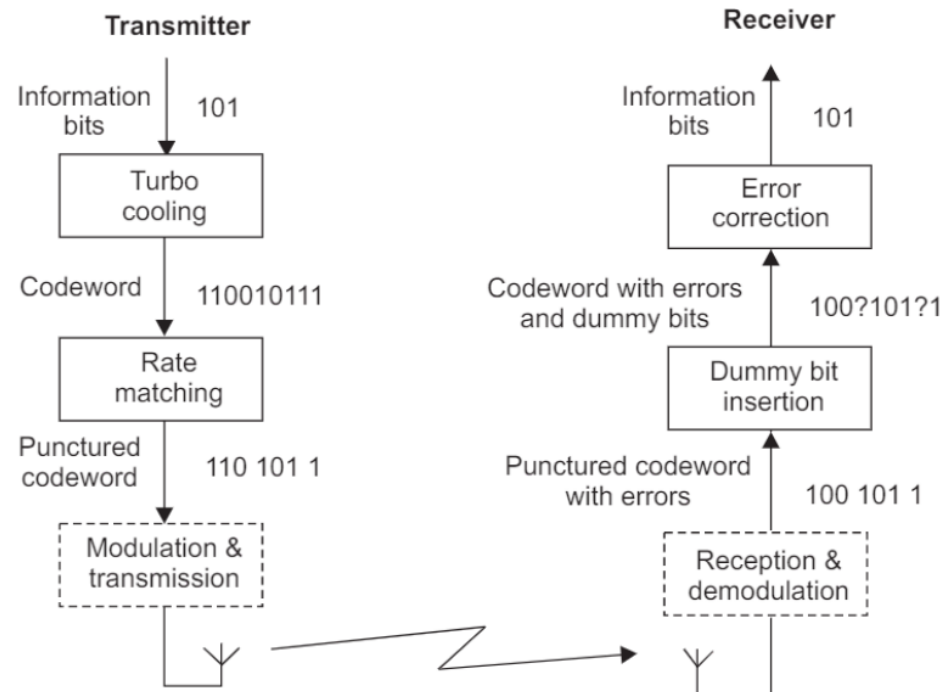
Error Management

- Noise and interference lead to errors in a wireless communication
- **Forward error correction** - technique used for controlling errors in data transmission over unreliable or noisy communication channels.
- the transmitted information is represented using a codeword that is typically two or three times as long.
- The extra bits supply additional, redundant data that allow the receiver to recover the original information sequence.

Forward Error Correction

- For example, a transmitter might represent the information sequence 101 using the codeword 110010111. After an error in the second bit, the receiver might recover the codeword 100010111. If the coding scheme has been well designed, then the receiver can conclude that this is not a valid codeword, and that the most likely transmitted codeword was 110010111.
- The coding rate - number of information bits divided by the number of transmitted bits ($1/3$ in the example above).
- forward error correction algorithms operate with a fixed coding rate. Despite this, a wireless transmitter can still adjust the coding rate using the two- stage process shown in the fig

Forward Error Correction



Block diagram of a transmitter and receiver using forward error correction and rate matching

Forward Error Correction

- The main algorithm used by LTE is known as Turbo coding and has a fixed coding rate of $1/3$.
- In the second stage, called rate matching, some of the coded bits are selected for transmission, while the others are discarded in a process known as **puncturing**.
- The receiver has a copy of the puncturing algorithm, so it can insert dummy bits at the points where information was discarded. It can then pass the result through a turbo decoder for error correction.

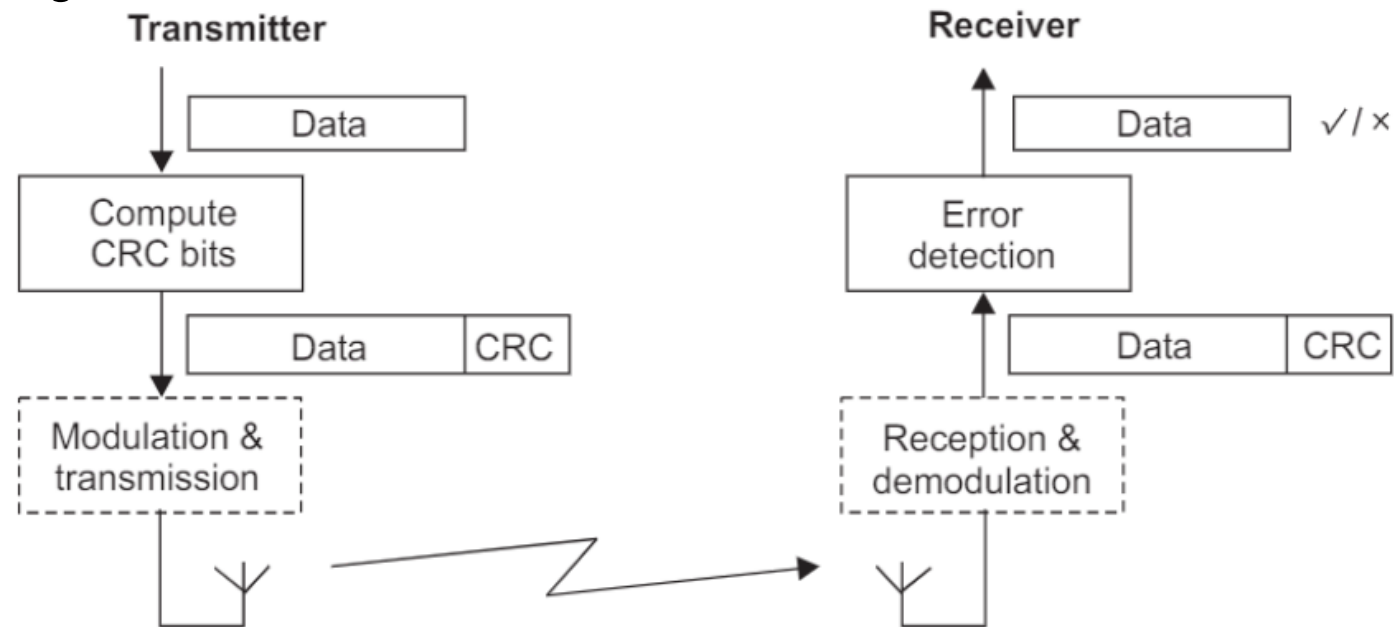
Forward Error Correction

- If the coding rate is low, the transmitted data contain many redundant bits which allows the receiver to correct a large number of errors and to operate successfully at a low SINR(Signal to Interference & Noise ratio), but at the expense of a low information rate.
- If the coding rate is close to 1, then the information rate is higher but the system is more vulnerable to errors.
- A trade-off between information rate and SINR has to be achieved.
- **Note : SINR(Signal to Interference & Noise ratio)** - the power of a certain signal of interest divided by the sum of the interference power (from all the other interfering signals) and the power of some background noise.

Automatic Repeat Request

Automatic Repeat Request

- Automatic repeat request (ARQ) is another error management technique, which is illustrated in Fig.



Block diagram of a transmitter and receiver using automatic repeat request

Automatic Repeat Request

- Transmitter takes a block of information bits and uses them to compute some extra bits that are known as a cyclic redundancy check (CRC).
- It appends these to the information block and then transmits the two sets of data in the usual way.
- Receiver separates the two fields and uses the information bits to compute the expected CRC bits.
- If the observed and the expected CRC bits are the same, then it concludes that the information has been received correctly and sends positive acknowledge back to the transmitter.

Automatic Repeat Request

- If CRC bits are different, then it concludes that the error has occurred and sends negative acknowledge back to the transmitter to request retransmission
- Positive and negative acknowledgements are often abbreviated to ACK and NACK respectively.
- A wireless communication system often combines these two error management techniques. Such a system corrects most of the bit errors by the use of forward error correction and then uses automatic repeat requests to handle the remaining errors that leak through.

Antennas

- Antenna is a device for converting electromagnetic radiation in space into electrical currents in conductors or vice-versa, depending on whether it is being used for receiving or for transmitting, respectively.
- Antennas transform wire propagated waves into space propagated waves
- Antennas receive electromagnetic waves and pass them onto a receiver or they transmit electromagnetic waves which have been produced by a transmitter.

Features of an Antenna

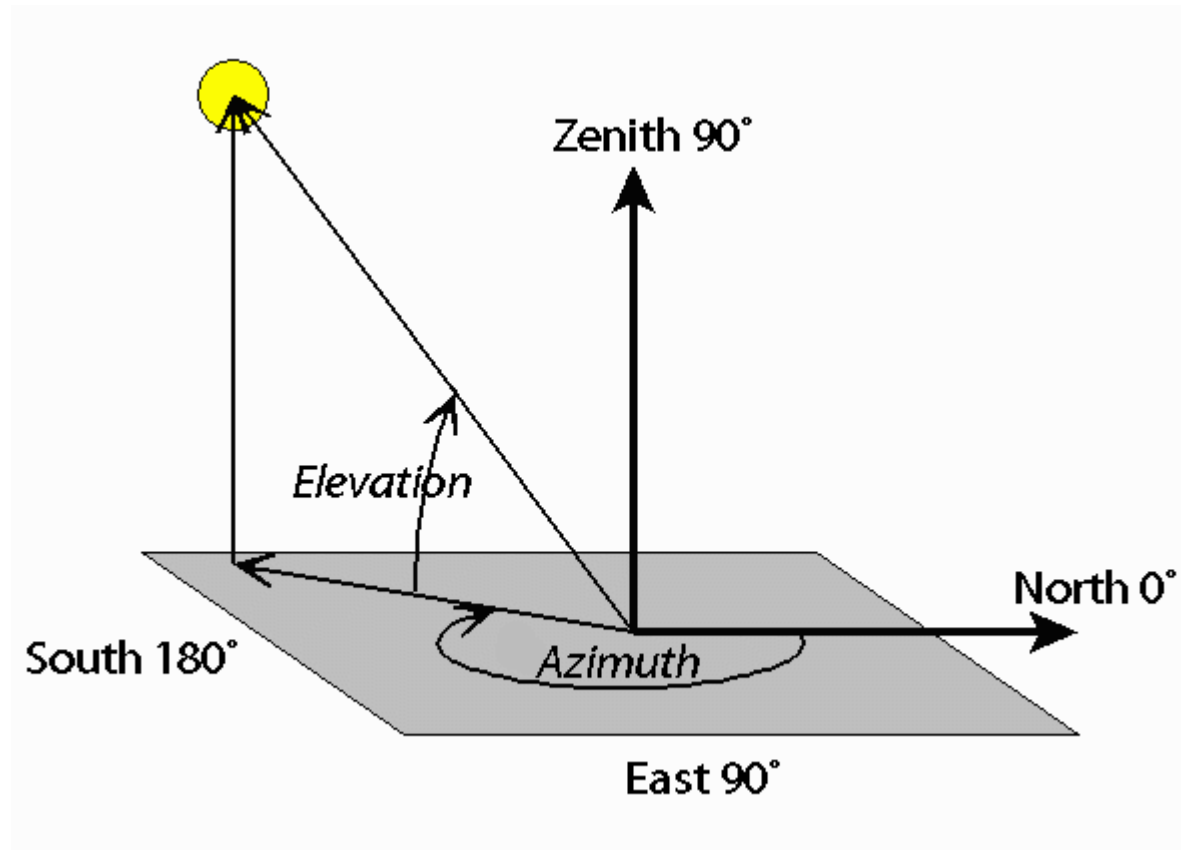
Following are the required features of an antenna

- Strictly defined radiation pattern for most accurate network planning
- Dual polarization
- Electrical down tilting of vertical diagram
- Unobstructive design

Some Common Antennas

- Omnidirectional Antennas
- Dipole Antennas
- Collinear omni Antennas
- Directional Antennas
- Patch Antennas
- Patch Array Antennas
- Yagi Antennas

Concept of Azimuth and elevation angle



Omnidirectional Antennas

- Non directional pattern (circular pattern) in a given plane
- Radiates equal power in all directions perpendicular to the axis



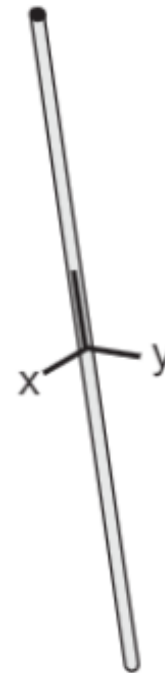
Radiates energy equally in all
horizontal directions

Examples of omnidirectional Antennas – Dipole Antennas and Collinear antennas

Dipole Antennas

- Most commonly referred as half-wavelength ($\lambda/2$) dipole
- The physical antenna is constructed of conductive elements whose combined length is half of the wavelength at its intended frequency of operation.

Dipole Antenna model



Collinear Omni Antennas

- To create an omnidirectional antenna with higher gain, multiple omnidirectional structures are arranged in linear or vertical fashion to retain the same omnidirectional pattern in the azimuth plane but a more focused elevation beam which leads to higher gain.
- Also frequently referred as collinear array
- Higher gain implies same power radiated in a more focused way



Directional Antennas

- Radiates energy most effectively in one direction than the others
- They have one main lobe and several minor lobes
- Used for coverage as well as point to point links
- They can be patch, dish or horn antennas
- They accomplish the main goal- radiating their energy in a particular direction

Examples of directional Antennas



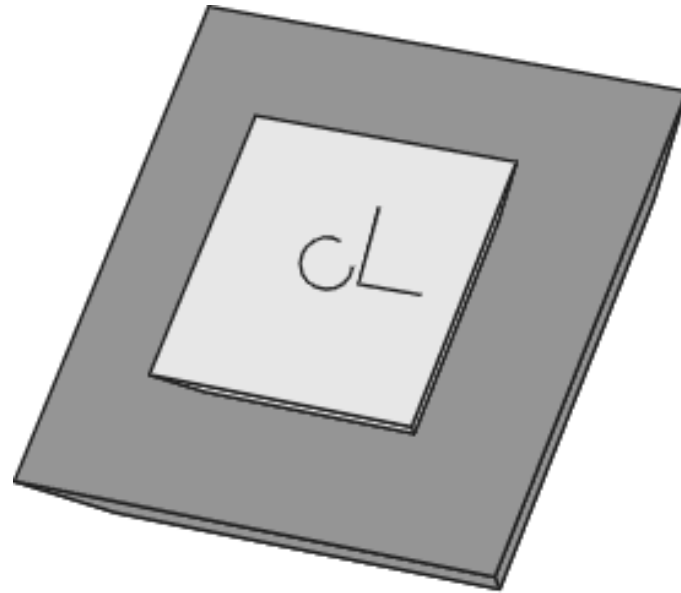
Dish Antenna



Horn Antenna

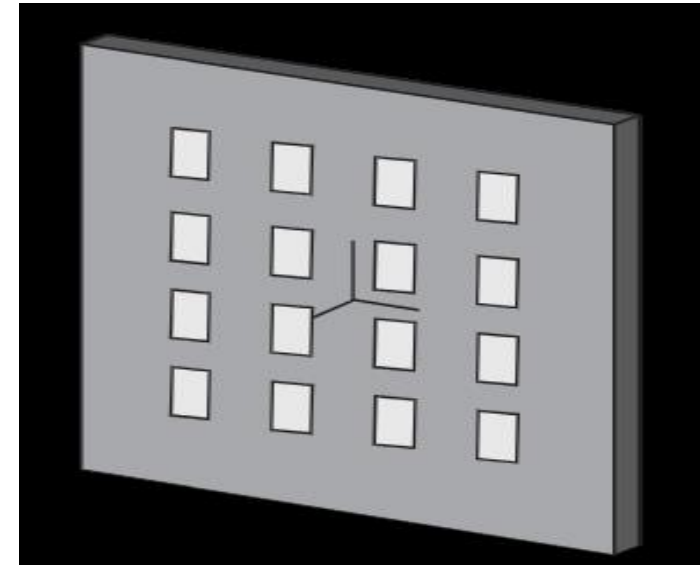
Patch Antennas

- In its simplest form is a single rectangular conductive plate that is spaced above a ground plane.
- Attractive due to their ease of fabrication



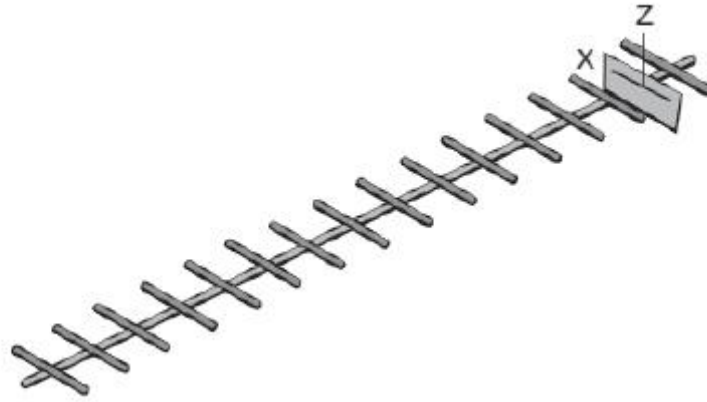
Patch Array Antennas

- Arrangement of multiple patch antennas that are all driven by the same source.
- Frequently this arrangement consists of patches arranged in orderly rows and columns
- Reason for this arrangement is higher gain



Yagi Antenna

- A Yagi antenna is formed by driving a simple antenna, typically a dipole or dipole-like antenna, and shaping the beam using a well-chosen series of elements whose length and spacing are tightly controlled.
- The Yagi shown in the figure below is built with one reflector (the bar behind the driven antenna) and 14 directors (the bars in front of the driven antenna).



Yagi Antenna

- Yagi antenna is a directional antenna that radiates its energy out in one main direction.
- Very often, these antennas are enclosed in a tube, with the result that the user may not see all the antenna elements.

Question Bank

1. With relevant waveforms define and explain amplitude modulation.
2. Define the following terms a) Frequency modulation b) Phase modulation
3. Explain the following with the help of waveforms
a) PAM b) PWM c) PPM d) PCM
4. With neat diagram explain different types of radio wave propagation.
5. What are the four modulation schemes used in LTE?
6. Explain the various Multiple Access Techniques used in cellular network.
7. Discuss the transmission modes used in mobile communication system.

Question Bank

8. Define the following

- i) Multipath
- ii) Constructive and destructive interference
- iii) Coherence time
- iv) Coherence bandwidth
- v) Delay spread

9. Write short notes on

- vi) Forward error correction
- vii) Automatic Repeat Request

10. List the important features of antennas.

11. Define Antennas and discuss the various common antennas used.